

Subsidy Pass-through and Market Power: Evidence from Production-Based Solar Subsidies

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Abstract

I examine the pass-through of solar subsidies in Massachusetts by comparing two distinct incentive programs that pay residential solar adopters based on the electricity their systems generate: the Solar Carve-Out program, which compensates owners through tradable SRECs sold at market prices, and its successor, SMART, which pays a fixed rate set at the time of application. A key contribution is the construction of subsidy measures specific to each project that account for heterogeneity in electricity generation potential, which allows me to identify pass-through from differences in generation across roofs. Baseline estimates are 76% under Solar Carve-Out and 102% under SMART, but much of this gap reflects how each subsidy moved over time alongside falling installation costs. Using only variation in generation across roofs, pass-through is about 85% in both programs, with no significant difference between them. Demand under SMART is log-concave, so market power would push pass-through below one. High pass-through therefore bounds installer markups at a few percent of price at the point estimates. Unlike the over-shifting found for leased systems in California, the evidence points to an installer market close to competitive.

1 Introduction

Government subsidies for renewable energy technologies have become a cornerstone of climate policy worldwide, with solar photovoltaic systems receiving particularly substantial support. In the United States alone, federal, state, and local governments have collectively spent billions of dollars subsidizing residential solar installations through various mechanisms including tax credits, rebates, and net metering incentives ([Borenstein, 2017](#)). The economic rationale for these subsidies rests on the premise that they will reduce the net cost of adoption for consumers, thereby accelerating the deployment of clean energy technologies and replacing fossil-fuel-based electricity generation, which comes with negative environmental externalities.

However, the effectiveness of these subsidies in reducing consumer costs depends critically on the rate at which they are passed through to buyers rather than being captured by suppliers. Economic theory predicts that subsidy incidence, which is the distribution of subsidy benefits between consumers and producers, depends on the relative elasticities of supply and demand, market structure, and the competitive environment. In markets with limited competition or significant market power, suppliers may capture a substantial portion of subsidy value through higher pre-subsidy prices, reducing the policy’s effectiveness in stimulating adoption while increasing its fiscal cost per unit of new capacity installed.

The empirical literature on subsidy pass-through in solar markets has grown substantially in recent years, though evidence remains concentrated in specific geographic contexts and program designs. [Pless and van Benthem \(2019\)](#) provide seminal evidence from California’s Solar Initiative (CSI), documenting a 78% pass-through rate for host-owned residential systems while finding more-than-complete pass-through exceeding 150% for third-party-owned systems. They argue that such over-shifting is consistent with imperfect competition combined with sufficiently convex demand curves, as predicted by the theoretical framework of [Weyl and Fabinger \(2013\)](#). [Gillingham and Tsvetanov \(2019\)](#) find an 84% pass-through rate for Connecticut’s solar rebate program using a structural demand estimation approach.

[Dong, Wiser and Rai \(2018\)](#) also examine the CSI program and report near-complete pass-through for residential installations. These studies collectively suggest that consumers generally capture a substantial share of subsidy benefits, with considerable variation depending on market structure and ownership arrangements.

Beyond solar-specific studies, the broader literature on subsidy and tax incidence provides important context. [Sallee \(2011\)](#) documents full pass-through of federal tax credits for the Toyota Prius despite binding production constraints, attributing this surprising result to Toyota’s strategic pricing decisions to cultivate the Prius brand. [Busse, Silva-Risso and Zettelmeyer \(2006\)](#) find that car buyers capture 70 to 90 percent of manufacturer cash rebates. [Cabral, Geruso and Mahoney \(2018\)](#) find incomplete pass-through of Medicare Advantage subsidies, with market concentration explaining much of the variation—premium pass-through was only 13% in the least competitive markets compared to 74% in the most competitive markets. [Genakos and Pagliero \(2022\)](#) likewise find that pass-through of fuel taxes on Greek islands rises from 0.4 in monopoly markets to 1 in markets with four or more competitors, and [Lade and Bushnell \(2019\)](#) find that half to three-quarters of the Renewable Fuel Standard subsidy for E85 reaches consumers, with retailers’ market structure influencing both the speed and the level of pass-through. Studies of tax and cost pass-through in other sectors find rates ranging from more than complete for alcohol taxes in Alaska ([Kenkel, 2005](#)) to high but incomplete for carbon emissions costs in electricity markets ([Fabra and Reguant, 2014](#)), and [Ganapati, Shapiro and Walker \(2020\)](#) use pass-through estimates to measure the incidence of energy costs in US manufacturing under imperfect competition.

A parallel strand of literature examines how subsidies affect solar adoption rates and how their design matters. [Hughes and Podolefsky \(2015\)](#) find that a \$0.10 per watt increase in CSI rebates increased daily installations by 7–15%, with larger effects during later program stages when rebates were smaller. [Crago and Chernyakhovskiy \(2017\)](#) use data from Northeastern states to show that \$1 per watt rebates increase annual adoption rates by approximately 50%. [De Groote and Verboven \(2019\)](#) estimate that Belgian consumers apply discount

rates averaging 15% to production-based solar subsidies—substantially higher than market interest rates—suggesting that the timing and structure of subsidy payments significantly affect their perceived value. [Burr \(2016\)](#) compares upfront versus production subsidies in the solar market and concludes that while upfront subsidies induce more adoption, production subsidies are more economically efficient.

Despite this rich literature, important gaps remain. First, most empirical evidence on solar subsidy pass-through comes from California’s CSI program, raising questions about external validity. Second, few studies have examined production-based certificate programs like Massachusetts’ Solar Carve-Out, which operate through tradable renewable energy certificates rather than direct rebates. Third, the literature lacks systematic comparisons of pass-through rates across different subsidy mechanisms within the same market, making it difficult to assess whether program design affects incidence. Fourth, prior studies have generally relied on ZIP-code-level solar resource estimates or assumed uniform capacity factors, potentially introducing measurement error in subsidy calculations and leaving unused the variation in subsidy value across roofs, which does not depend on the policy calendar. Finally, pass-through has been used to detect market power in solar markets ([Pless and van Benthem, 2019](#)), but not to bound it where pass-through is high.

This paper addresses these gaps by examining subsidy pass-through in Massachusetts, where two distinct production-based incentive programs created variation in both the magnitude and structure of solar subsidies over time. The Solar Carve-Out II program (2014–2018) provided incentives through tradable Solar Renewable Energy Certificates (SRECs), with market prices that varied substantially across vintage years and generally tracked close to regulatory price ceilings. The program was replaced in late 2018 by the Solar Massachusetts Renewable Target (SMART) program, which offers fixed incentive rates that decline predictably according to predetermined capacity block schedules. These two programs differ not only in their incentive structures but also in their generosity, with average Solar Carve-Out subsidies approximately 3.4 times as large as SMART subsidies on a per-kilowatt basis

(\$1,923 versus \$566 in present value).

Using comprehensive installation-level data covering over 39,000 residential solar systems installed between 2014 and 2023, I estimate subsidy pass-through rates by examining how pre-subsidy installation prices respond to variation in subsidy generosity. A key methodological contribution is the construction of project-specific subsidy measures that account for heterogeneity in electricity generation potential. I merge detailed technical specifications from Lawrence Berkeley National Lab’s Tracking the Sun database with Massachusetts regulatory data, then use the National Renewable Energy Laboratory’s PVWatts calculator to estimate annual generation for each installation based on its precise geographic coordinates, system size, panel orientation, and equipment characteristics. This approach allows me to calculate the net present value of production-based subsidies for each roof, rather than from ZIP-code-level averages or assumed uniform capacity factors, and the resulting differences in subsidy value across roofs in the same town, installed by the same installer in the same month, are what identify pass-through in my preferred specification.

My empirical strategy exploits two sources of variation in subsidy levels: changes in the subsidy over time, from the market prices of SRECs under Solar Carve-Out and the declining-block structure of SMART incentive rates, and differences in expected generation across roofs, which exist in both programs and do not depend on the policy calendar. I estimate pass-through by regressing net installation costs on subsidy amounts while controlling for system quality through inverter and panel manufacturer fixed effects, local market conditions through city fixed effects, installer fixed effects, and time trends through year fixed effects. For the SMART program, I also include utility fixed effects to account for differences in compensation rate schedules across the three investor-owned utilities that administer it. To isolate the variation across roofs, I then replace year fixed effects with year-month fixed effects, which absorb the SREC price path and the timing of SMART block changes, and add installer-by-year and city-by-year fixed effects. I cluster standard errors at the installer level.

I find that baseline subsidy pass-through rates differ between the two programs. For Solar Carve-Out installations, the estimated pass-through rate is 76%, indicating that for each dollar of subsidy provided, consumers pay 76 cents less in net installation costs. This estimate is close to prior estimates for solar rebate programs despite differences in program design and geographic context, and it is statistically different from one. For SMART installations, I estimate pass-through of 102%, which cannot be distinguished from complete pass-through ($p = 0.80$). Much of this gap, however, comes from how each subsidy moved over time relative to installation costs. When the timing of the subsidy is absorbed and only differences in generation across roofs identify pass-through, the Solar Carve-Out estimate rises to 0.86 and the SMART estimate falls to 0.85–0.89, and the two programs are no longer statistically different. Several SMART estimates that rely on timing exceed one—1.55 and 1.32 in the two lowest quartiles of the subsidy and 1.27 at the step-downs in SMART rates—which log-concave demand cannot produce unless marginal cost falls with volume, so they point to bias from the timing variation rather than to over-shifting. I therefore take pass-through of about 85% in both programs as the main result. The Solar Carve-Out estimates are robust to alternative methods for calculating expected subsidy values and are below one at every discount rate. Two alternative designs based on discrete changes in the subsidy—the January 2017 cut in the SREC factor and the step-downs in SMART rates as capacity blocks fill—give estimates of similar magnitude, though with less precision.

I then ask what high pass-through implies about market power, using the sufficient-statistics framework of [Weyl and Fabinger \(2013\)](#) as [Pless and van Benthem \(2019\)](#) do, but in reverse. Pless and van Benthem find more-than-complete pass-through together with convex demand and conclude that solar markets in California are imperfectly competitive. Following them, I estimate demand at the ZIP-code level and find that demand under SMART is log-concave. With log-concave demand, market power would push pass-through below one: at the point estimates, installer markups are at most about 2–4% of price, and baseline pass-through near one implies a nearly competitive market with close to flat marginal cost. The

bound is wider at the upper end of the confidence region, and demand under Solar Carve-Out is not identified, so this conclusion applies to SMART. Reduced-form checks agree: pass-through does not fall when installers are busy and does not vary significantly with local installer concentration.

Beyond the timing of the subsidy, I consider three explanations for the difference in baseline pass-through between the programs: SMART subsidies are substantially smaller, SMART rates are more predictable, and the competitive environment may have differed between the two periods. I find little support for the third explanation: statewide installer concentration is low throughout, the top installers' market share falls mainly before SMART begins, and SMART pass-through does not vary significantly with local concentration. The first two explanations are consistent with the data but cannot be cleanly separated from each other, because the two programs differ in size, certainty, and time period at once, and the similar cross-roof estimates suggest that neither is large. I emphasize what the data cannot distinguish.

The remainder of the paper proceeds as follows. Section 2 provides institutional background on the Solar Carve-Out and SMART programs. Section 3 describes the data sources and construction of key variables, and explains the methodology for estimating annual electricity generation and calculating subsidy net present values. Section 4 presents the empirical framework and maps each source of identifying variation to the specification that isolates it. Section 5 reports the baseline estimates, the estimates from cross-roof variation, two alternative designs that use discrete changes in the subsidy, and robustness to the treatment of the price level, SREC price assumptions, and discount rates. Section 6 estimates demand and uses it to bound installer market power. Section 7 evaluates the three candidate explanations for the difference in baseline pass-through between programs. Section 8 concludes with implications for policy design. The appendix reports additional specifications, including details of the alternative designs, heterogeneity by income and installer concentration, and alternative demand specifications.

2 Institutional Background

When an individual in Massachusetts installs a solar system, one of the two types of production-based subsidies is available depending on the policy regime at the time of installation. If the qualified system entered commercial operation between 2008 and November 2018, the owner would be qualified to receive certificates under the Solar Carve-Out program. If a system entered commercial operation after November 2018, the owner would be entitled to receive an incentive rate on top of net metering through their electricity bill under the SMART program. To explain how subsidies are calculated, I first discuss the two incentive programs and then introduce the data and method I use for the calculation.

2.1 Solar Carve-Out

The Solar Carve-Out is part of the Renewable Portfolio Standard, which mandates that a certain percentage of electricity come from solar resources. Launched on January 1, 2010, the program's requirement increased over time until reaching a capacity target of 400 MW. The state government expanded the program by launching Solar Carve-Out II in April 2014, with a total goal of 1,600 MW in solar PV capacity.

Energy suppliers, including utilities and competitive energy suppliers, are required to purchase Solar Renewable Energy Certificates (SRECs) to demonstrate compliance or make an alternative compliance payment (ACP). NEPOOL GIS tracks solar generation and issues SRECs quarterly to solar generation unit owners at a rate of 0.45–1.0 SREC per MWh, depending on system characteristics (the SREC factor). For example, a system smaller than 25 kW authorized to interconnect before January 8, 2017, receives a factor of 1.0. Each qualified solar generation unit is entitled to receive SRECs for 40 calendar quarters as long as the unit remains operational. After 40 calendar quarters, the system generates Renewable Energy Certificates (RECs) instead, which are arguably much less valuable than SRECs.

Under the Massachusetts SREC program, the vintage of renewable energy certificates

is determined by the calendar year in which the electricity is generated, with each year assigned a distinct SREC product designation. For instance, all electricity generated by a solar installation during the 2014 calendar year entitles the system owner to MA2014(II) certificates, where the prefix “MA” denotes Massachusetts, “2014” indicates the generation year, and the Roman numeral “II” designates the applicable program phase—in this case, Solar Carve-Out II. The annual vintage structure creates distinct market segments, as SRECs from different years may trade at varying prices depending on factors such as program caps, alternative compliance payment levels, and the balance between supply and demand within each vintage year.

Solar generation unit owners can either market and trade SRECs independently or join an aggregation that represents solar owners in harvesting and trading SRECs. Trading platforms and aggregators typically charge fees ranging from 1–10% of the proceeds depending on system size, with small residential systems (under 50 kW) commonly paying 10%. DOER holds an annual fixed-price clearinghouse auction from May 16 to June 15 to settle SRECs not traded during the previous year. If certificates remain unsold through the auction, owners can still trade them in the market for the next two compliance years. The ACP provides a price ceiling for SRECs, while the clearinghouse auction establishes a price floor. In the Solar Carve-Out’s launch year, the ACP was set at \$600/MWh, while the net clearinghouse auction fixed price was \$285/MWh (a \$300 price less the 5% auction fee). This mechanism guaranteed owners a minimum return of \$285/MWh, if sold through the auction. In practice, market prices for SRECs tracked closer to the ACP rate, resulting in returns exceeding twice the electricity price in 2010. For Solar Carve-Out II the ACP and auction price schedules were published at launch, whereas under the original program the ACP was adjusted over time. This narrowed the range of possible SREC prices but did not remove price risk: as Figure 1 shows, market prices sat below the auction floor for several years.

Figure 1 shows the historical market prices for SREC IIs alongside ACP and clearinghouse auction prices. In the first few years, market prices were consistently below auction prices

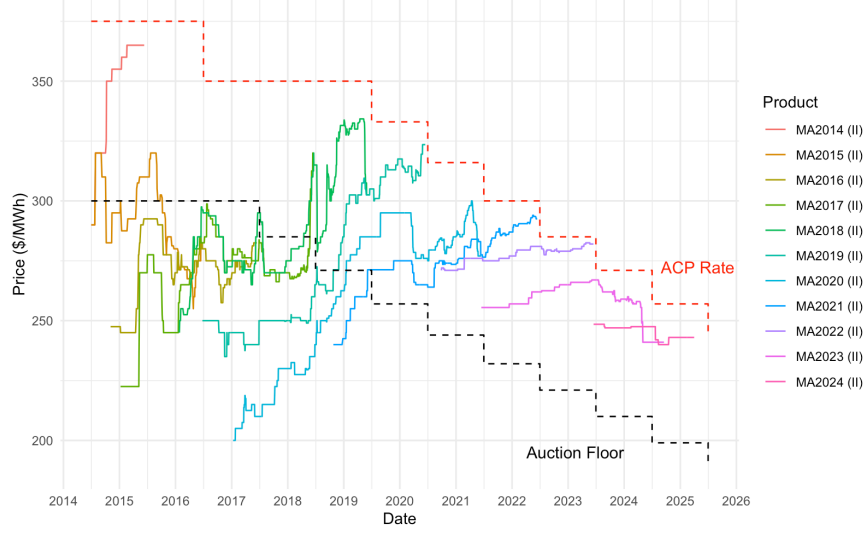


Figure 1: SREC II Market Prices with ACP and Auction Price Bounds. The auction floor is plotted gross of the 5% auction fee. An owner selling through the auction receives 95% of the plotted value (\$285/MWh in 2014–2016).

due to an oversupply in the market. In later years, market prices for SRECs followed the declining trend of the ACP. The net clearinghouse price remained constant at \$285/MWh from 2014 to 2016, then decreased at a rate of 5% per year. The net clearinghouse price stood at \$189.05/MWh in 2024.

2.2 SMART

The Solar Massachusetts Renewable Target (SMART) is a program created by the Department of Energy Resources (DOER) to further incentivize solar development in Massachusetts after the Solar Carve-Out II target was met. The program was officially launched on November 26, 2018. Under this regulation, the three investor-owned utilities (IOUs) in Massachusetts: Eversource, National Grid, and Unitil, are required to compensate solar system owners based on the amount of electricity generated, in exchange for the renewable energy certificates (RECs) attached to this generation to be transferred to them.

The actual incentive rate is determined by four factors: the base compensation rate, household income status, whether the system includes a storage component, and the value

of energy. The base compensation rate starts high and declines according to a predetermined schedule as capacity blocks are filled, with both capacity block sizes and rate schedules varying by utility. The value of energy is set from 3-year average retail electricity prices (delivery and transmission charges included) at the time of application. The SMART incentive rate is calculated by subtracting the value of energy from the base compensation rate. Small residential solar generation unit owners receive payments through the utility equal to this rate multiplied by their monthly generation, in addition to the electricity bill savings already realized through their behind-the-meter system, and they are able to cash out the negative balance monthly. As electricity prices rise and capacity blocks advance, the SMART incentive rate decreases until it reaches zero¹.

Table 1 shows the base compensation rates for generation units of 25 kW AC or less, by utility. There are eight blocks for the three large utilities but only two and four for Nantucket and Fitchburg, respectively. Fitchburg is owned by Unitil, while Mass Electric and Nantucket Electric are National Grid’s operating companies in Massachusetts.

Table 1: SMART Compensation Rates for Generation Units Less Than or Equal to 25 kW AC

Utility	Low income	Block 1	2	3	4	5	6	7	8
Fitchburg	Y	\$0.358	\$0.326	\$0.298	\$0.272	—	—	—	—
	X	\$0.311	\$0.284	\$0.259	\$0.236	—	—	—	—
Mass Electric	Y	\$0.358	\$0.344	\$0.330	\$0.317	\$0.304	\$0.292	\$0.280	\$0.269
	X	\$0.311	\$0.299	\$0.287	\$0.275	\$0.264	\$0.254	\$0.244	\$0.234
Nantucket	Y	\$0.391	\$0.328	—	—	—	—	—	—
	X	\$0.340	\$0.286	—	—	—	—	—	—
Eversource E.	Y	\$0.391	\$0.375	\$0.360	\$0.346	\$0.332	\$0.319	\$0.306	\$0.294
	X	\$0.340	\$0.326	\$0.313	\$0.301	\$0.289	\$0.277	\$0.266	\$0.255
Eversource W.	Y	\$0.329	\$0.315	\$0.303	\$0.291	\$0.279	\$0.268	\$0.257	\$0.247
	X	\$0.286	\$0.274	\$0.263	\$0.253	\$0.243	\$0.233	\$0.224	\$0.215

Base compensation rates in dollars per kWh by utility and capacity block.

The SMART program is favorable early in the program life cycle because the incentive

¹System owners can also participate directly in the RECs market if they do not wish to enroll in the SMART program, and they are advised to do so by DOER when the SMART rate drops to a low level.

rate is considerably higher than the price of RECs. In my sample the SMART incentive rate reaches \$0.20/kWh at most, with a median of about \$0.066/kWh, when RECs were trading at around \$0.04/kWh². SMART also offers more predictable future cash flows than the Solar Carve-Out program, as the rate does not fluctuate with market conditions and remains constant over time. This allows for precise subsidy estimates based on generation projections. Another key difference between SMART and Solar Carve-Out is that SMART requires system interconnection to one of the three IOUs. As a result, households served by municipal utilities are ineligible for the program.

2.3 Comparison of the Two Programs

Table 2 summarizes the two programs. Four differences matter for how subsidy value could be divided between buyers and installers. First, the form of payment: an owner under Solar Carve-Out sells certificates whose price is set in a market bounded by the ACP and the auction floor, whereas a SMART owner receives a fixed per-kWh rate. Second, the timing of information: the SMART rate is fixed when the application is approved, so the buyer and installer know the subsidy per kWh for ten years, while the SREC price the owner will eventually receive is unknown at the time of purchase. Third, the size of the subsidy: the mean present value is about \$1,923 per kW under Solar Carve-Out and \$566 per kW under SMART, roughly 45% and 15% of the average gross price, with almost no overlap between the two distributions. Fourth, coverage: Solar Carve-Out is open to all utility territories, while SMART is limited to customers of the three IOUs.

These differences generate competing predictions. A larger subsidy that arrives through a volatile market price is harder to price into a quote and may be more likely to strain installers' capacity, which would lower pass-through. A small, certain subsidy is easy to incorporate into a price and is less likely to move installer capacity, which would raise pass-through. Because the programs also differ in years, technology costs, and eligible households,

²Source: SREctrade trading platform

Table 2: Comparison of the Solar Carve-Out II and SMART Programs

	Solar Carve-Out II	SMART
Period in sample	01.2014 – 11.2018	03.2019 – 09.2023
Form of payment	Tradable SRECs	Fixed rate per kWh
Rate	Market price with regulation	Fixed at approval
Payment term	40 calendar quarters	10 years
Payer	Retail electricity suppliers	The three IOUs
Eligibility	All utility territories	IOU customers only
Mean subsidy (NPV, 8%)	\$1,923 per kW	\$566 per kW
Share of cost	~45%	~15%

Subsidy is the present value per kW of DC capacity at an 8% discount rate with 0.5% annual degradation, from the analysis samples. Shares are ratios of the sample mean subsidy to the sample mean gross price per kW (systems of 25 kW or less).

I treat these as hypotheses to be examined in Section 7, not as identified effects.

3 Data

3.1 Data Sources and Sample

I obtain the solar installation data from the Massachusetts Department of Energy Resources (DOER). For SMART projects, the DOER data report compensation rates as well as project characteristics including location, utility, size, installation cost, installer, and technical details. These technical details, which include azimuth, tilt, and inverter rating, are crucial for accurate solar generation estimation. The compensation rate reported in the dataset is the actual SMART incentive rate that system owners will receive, which reflected various adders and value of energy.

For Solar Carve-Out II projects, DOER reports similar variables but without technical details. To obtain annual generation estimates as accurate as those for SMART, I merged the DOER dataset with the Tracking the Sun database made available by Lawrence Berkeley National Lab (LBNL). Tracking the Sun is a cleaned dataset covering over 60% of the solar projects in the United States. LBNL cleans and consolidates information about solar projects from individual states' energy departments on a yearly basis. The Massachusetts DOER and Clean Energy Center are both prominent data providers for LBNL, hence all projects in my dataset can be found in Tracking the Sun, along with their technical details. One challenge in merging the datasets was that projects are identified differently in the two sources. To address this, I used a merge criterion that matches operation date, size, and total cost simultaneously. This resulted in an almost perfect one-to-one merge, with only a few projects matched to the same Tracking the Sun record. I manually verified that these projects indeed have the same technical details. The SREC prices are not available in the project dataset. I constructed a SREC price dataset using information from a major SREC trading platform SRECTrade and price support regulations published in DOER's website.

The SREC sample spans from January 1, 2014, to November 26, 2018, the period when Solar Carve-Out II was open to applications. The SMART sample begins on March 25,

2019, marking the debut of the incentive program, and ends on September 20, 2023, when most projects received zero compensation and the number of applications dropped sharply. For the SMART sample, the DOER data list two dates for each project: the Commercial Operation Date and the Incentive Payment Effective Date. I use the Commercial Operation Date, as this is when compensation rates were determined.

I include only host-owned, small-scale (≤ 25 kW DC) residential projects without battery installations in the sample, for the following reasons. First, the general net metering program has a capacity cap, which means projects installed after the cap was met might not be eligible to participate in the net metering program. Small-scale systems are exempt from the cap, so including only small-scale projects ensures that all systems in the sample are on net metering. Second, only host-owned projects have directly observed prices. Third-party-owned projects are said to report the “market” value to DOER, which is not the true cost to system owners ([Pless and van Benthem, 2019](#)). Lastly, installations with battery storage qualify for a storage adder, which provides a higher incentive, and storage components would distort the cost of a solar system. I also remove observations with a per-kW price above the 99th percentile or below the 1st percentile to exclude outliers.

3.2 Annual Generation Estimation

To estimate subsidy pass-through, it is necessary to obtain the post-incentive cost for each project, which is essentially the cost reported in the data minus the subsidy. Because both the SMART and Solar Carve-Out II programs are production-based subsidies, the total subsidy depends on how much electricity a system will generate and when the payments are made. For the SMART program, payment is made every month through the utility company, so the subsidy can be computed by multiplying the incentive rate by monthly electricity generation. For the SREC program, payment comes from selling certificates, which are issued every three months, so the subsidy is the market price of certificates multiplied by the number of certificates produced per quarter. For simplicity, I calculate the subsidy on a

yearly basis, which requires an annual generation estimate for each project.

I estimate annual generation using the PVWatts calculator, a tool developed by the National Renewable Energy Laboratory (now the National Laboratory of the Rockies) for estimating the energy production of grid-connected photovoltaic systems. The calculator links a project’s location to its underlying solar irradiance and meteorological data, along with a set of system parameters, to estimate monthly generation.

The most important inputs for the calculator are a project’s size and location, as they represent the area of solar panels and the available solar resources for the project. The location input accepts street addresses, ZIP codes, or coordinates. The publicly available DOER dataset includes only ZIP codes for each project. However, because ZIP codes can represent very large areas while the underlying solar resource data are much more granular, using ZIP codes alone would introduce bias.

To obtain the most accurate possible estimates, I acquired detailed addresses for each project through a separate data request to DOER. When I used street addresses directly as the input, the calculator responded poorly. To solve this issue, I first converted the addresses into coordinates using the U.S. Census geocoder. For addresses that failed to geocode, I used the OpenStreetMap geocoding service instead. All addresses were successfully geocoded using these two services. I also verified that all coordinates were located in Massachusetts to avoid errors introduced during geocoding. Figure 2 shows the location of each project in the SMART (panel a) and Solar Carve-Out (panel b) samples. The main difference between SMART and Solar Carve-Out II is coverage, driven by the ineligibility of households in municipal utility territories for SMART. In both panels, the density of projects closely resembles the population distribution of Massachusetts.

Besides size and location, other required parameters include the DC-to-AC ratio, module type, array type, azimuth, and tilt. I use the coordinates as inputs in the PVWatts calculator. While system size is available in the data for every project, other parameters are not available for all projects. For these missing parameters, I calculate an implied value. For example,

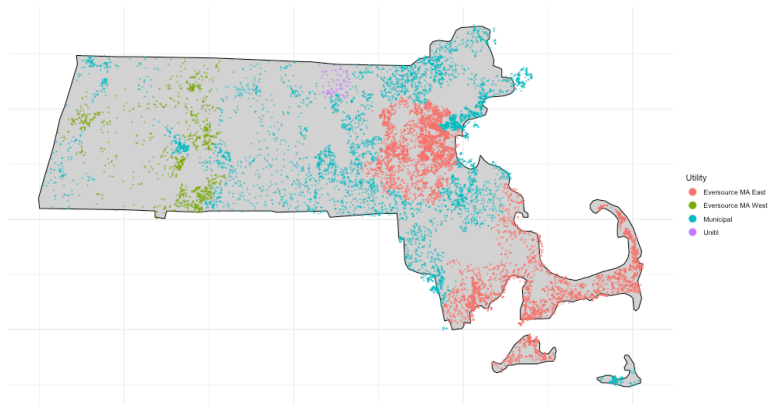
the DC-to-AC ratio in the original SMART dataset is calculated using the DC and AC ratings. When the implied value is also unavailable, I use the default values provided by the calculator, which represent typical values for rooftop solar projects. Very few projects (fewer than 100) require default values.

The calculator output includes monthly and annual generation for each project. I calculate the capacity factor for each project using the formula $CF = \text{Annual Generation} \div (\text{Size} \times 24 \times 365)$, where 24 is the number of hours in a day and 365 the number of days in a year. The average estimated capacity factor is 0.1429, or approximately 14.29%. The value is higher than the capacity factor reported by DOER (13.1%) based on actual generation data for 2010–2023, partly because the state’s solar portfolio consists of a mix of old and new solar panels. Panel aging can lower performance. Section 3.3 examines this gap. Most projects are in the 12.5–15% range, while some display a very low capacity factor. By manually entering the technical specifications into the calculator, I verified the accuracy of these estimates. The primary reason for a low capacity factor is suboptimal configuration. For example, a perfectly north-facing system will generate much less than a south-facing system, all else equal.

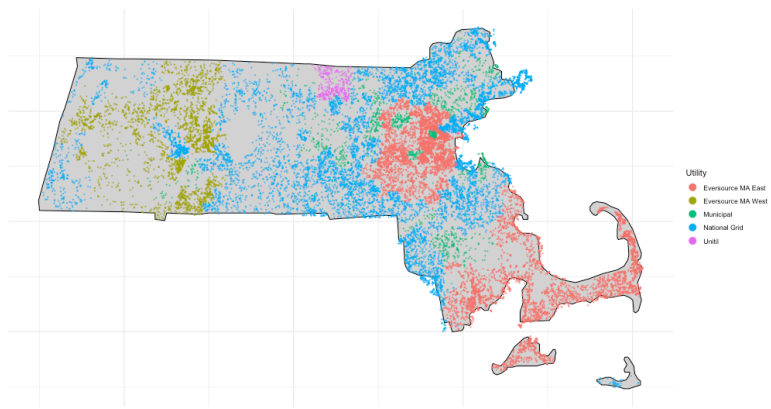
3.3 Validating the Generation Estimates

The subsidy is proportional to estimated generation, so the credibility of the estimates matters for both the regressor and the dependent variable. PVWatts is a modeled estimate, not a measurement. I fix the system loss parameter at the calculator default of 14.08%, assume a fixed roof-mount array with standard modules, and use observed size, tilt, azimuth, and DC-to-AC ratio where available. The calculator does not model site-specific loss from shading, snow cover, soiling and unavailability beyond the default value.

The average estimated capacity factor is 14.3%, above the 13.1% the Department of Energy Resources reports from metered generation for the whole Massachusetts fleet in 2010–2023. Panel degradation explains only part of this gap. Field evidence suggests that



(a) SMART



(b) Solar Carve-Out II

Figure 2: Project Locations by Utility Territory. Some dots fall outside the Massachusetts border because the Massachusetts shapefile has reduced precision when plotting.

panels lose about 0.5% of output per year (Deline et al., 2024), and applying that rate over the age distribution of the fleet closes roughly a fifth of the difference. The larger part of the gap is most plausibly losses that PVWatts does not model, including the default snow loss of 0%, which is unlikely to be the case in Massachusetts, and the default 3% shading and 2% soiling losses, which are likely underestimates considering the prolonged pollen season and tree removal restrictions. There are also lower-quality installations in the statewide fleet that pre-date my sample.

What matters for the pass-through estimates is not the level of generation but how well it ranks projects. Figure 3 shows that project-level capacity factors carry information that a town or statewide average would miss. The standard deviation across projects is 0.012 around a mean of 0.143, so more than half of projects (53%) differ from the statewide mean by more than 5% and 18% differ by more than 10%. About 91% of the variance is within towns, which means that assigning each project its town average would discard most of the variation in subsidy value across roofs.

Errors in the generation estimates affect the pass-through estimate in a known direction. Write the gross price as $P_i = c_i + (1 - \rho)S_i$ and the measured subsidy as $\hat{S}_i = S_i + u_i$. The dependent variable is the net price $P_i - \hat{S}_i$, so the error enters both sides of the regression, and with classical error the estimated pass-through is

$$\hat{\rho} = \lambda\rho + (1 - \lambda), \quad \lambda = \frac{\text{Var}(S)}{\text{Var}(S) + \text{Var}(u)}.$$

Classical measurement error therefore pulls the estimate toward one, not toward zero. Equivalently, in a regression of the gross price on the measured subsidy, the coefficient $1 - \rho$ is attenuated toward zero. Errors in the inputs, such as a mis-recorded tilt or azimuth or a geocoding error, are of this kind. Losses that PVWatts does not model and that are unrelated to its inputs, such as shading from a neighbor’s tree, are not: if true generation equals the estimate plus an independent error, the estimate is unbiased. The pull toward

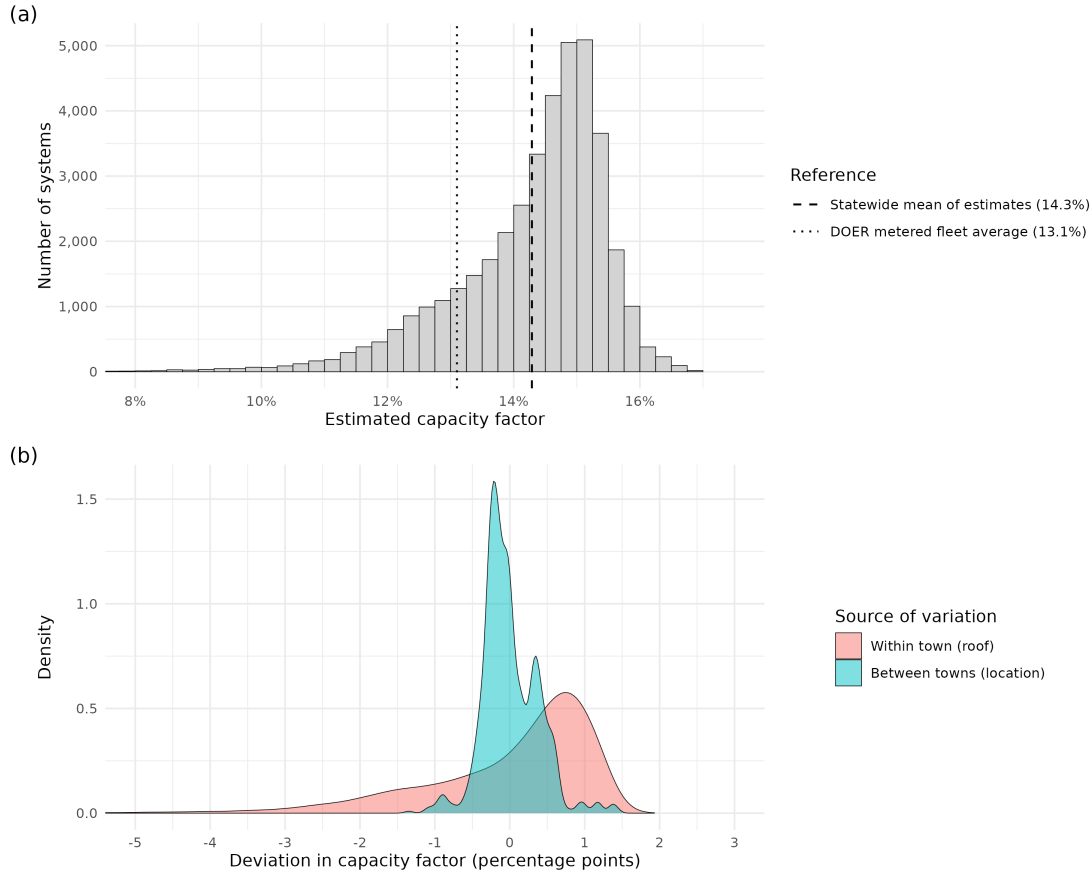


Figure 3: Variation in Estimated Capacity Factor Across Projects

one matters most when the identifying variation is narrow. Within the same installer, town, and month, roofs differ less in true generation than across the whole sample, so the share of error in the remaining variation is larger in the cross-roof specifications of Section 5.2, and instrumenting the subsidy with the estimated capacity factor does not remove it, because the instrument comes from the same PVWatts run.

3.4 Subsidy Calculation

I first calculate the net present value of the subsidy for SMART projects. I assume a discount rate of 8%. Because SMART compensation rate was locked in for 10 years at the time of application, the system owner will receive compensation based solely on annual solar panel generation in a given year. [De Groote and Verboven \(2019\)](#) suggest that system

owners significantly discount future cash flows generated by such subsidy programs, so I also conduct sensitivity tests using different discount rates. I follow a standard discounted cash flow model:

$$\text{NPV}_i = \sum_{t=1}^{10} \frac{r_i \times g_i (1-a)^t}{(1+d)^t}$$

where r_i is the SMART incentive rate, g_i is the estimated annual generation, d is the discount rate, and a is the solar degradation rate. I assume an annual solar degradation rate of 0.5% to account for the performance loss resulting from panel aging. Because r_i and g_i do not change over the ten years, the formula reduces to $\text{NPV}_i = r_i g_i A(d)$, where $A(d) = \sum_{t=1}^{10} (1-a)^t / (1+d)^t$ is the same for every project. The discount rate therefore rescales every SMART subsidy by the same factor, which matters for interpreting the robustness checks in Section 5.4.

Because SREC compensation rates were market-determined, they are not listed in the Solar Carve-Out data, and the price an owner would receive was not known at the time of purchase. In the initial years, market prices for SRECs were well below the ACP and even consistently below the auction price, so using either the ACP or the auction price alone to proxy for the compensation rate would be undesirable. In the baseline, I use the midpoint of the ACP and the auction price in each year, which is the price a buyer would expect without a view on where the market would settle within the published bounds. As an alternative, I proxy SREC compensation rates using the annual average market price of SRECs, which reflects the compensation owners actually receive in the future under the assumption of rational expectations. For projects in Solar Carve-Out II that were still receiving SRECs as of 2025³, this alternative uses the average 2024 price for the rest of their lifespan. Section 5.4 compares these and other price assumptions.

I apply the same approach to Solar Carve-Out projects but with said certificate prices:

$$\text{NPV}_i = \sum_{t=1}^{10} \frac{(g_i \div 1000) \times f_i \times 90\% \times P_t (1-a)^t}{(1+d)^t}$$

³Projects with an operation date after 2016 are still receiving SRECs, but the market prices have yet to be determined. For example, a 2018 project will keep receiving and monetizing SRECs until 2028, but SREC prices for 2026 through 2028 are still unknown.

where g_i is the estimated annual generation, f_i is the project’s SREC factor, d is the discount rate, a is the solar degradation rate, P_t is the SREC price in year t , and the 90% term reflects the aggregator fee. I discount the ten years of payments after commercial operation for both programs, and I use the midpoint of the ACP and the auction price as the baseline price path (Section 5.4). The Solar Carve-Out entitlement is 40 quarters, so ten years of payments covers the whole term.

3.5 Summary Statistics

Table 3 presents summary statistics for solar installations enrolled under the SMART program (Panel A) and the Solar Carve-Out program (Panel B), revealing substantial differences in incentive structures between the two policy mechanisms. The most striking disparity emerges in the subsidy per kW, which serves as a key indicator of program generosity and policy design. Solar Carve-Out installations received an average subsidy of \$1,922.77 per kW, approximately 3.4 times the mean SMART subsidy of \$565.60 per kW. As a share of the average gross price per kW, the subsidy is about 45% under Solar Carve-Out and 15% under SMART. This differential reflects the design difference between the two programs: Solar Carve-Out certificates were priced by a market that tracked a high ACP, whereas SMART rates fall with the value of energy and as capacity blocks fill.

The distribution of subsidies also differs. In absolute terms the dispersion is similar (a standard deviation of \$336 per kW for Solar Carve-Out and \$318 per kW for SMART), but relative to the mean it is much larger under SMART (a coefficient of variation of 0.56 compared with 0.17). Every Solar Carve-Out system received at least \$757 per kW, while some SMART systems received nothing because the value of energy had caught up with the base rate. The maximum subsidy is \$2,632 per kW under Solar Carve-Out and \$1,595 per kW under SMART, so the two distributions barely overlap. This matters for interpretation, because a comparison of the two programs is a comparison of very different parts of the subsidy range.

Table 3: Summary Statistics for SMART and Solar Carve-Out Samples

Variable	N	Mean	Std. Dev.	Min	Max
<i>Panel A: SMART Sample</i>					
Size (kW DC)	14,317	9.22	3.53	1.28	24.96
AC (kW)	14,317	7.49	2.64	1.20	23.50
Total Price (\$)	14,317	34,305.30	14,175.21	5,517.00	122,655.00
Price per kW (\$/kW)	14,317	3,795.74	881.11	1,998.32	7,008.21
Total Subsidy (\$)	14,317	5,182.54	3,705.16	0.00	31,917.29
Subsidy per kW (\$/kW)	14,317	565.60	317.63	0.00	1,594.76
Net Price (\$)	14,317	29,122.77	13,216.33	3,118.39	111,983.60
Net Price per kW (\$/kW)	14,317	3,230.14	970.62	834.28	6,976.19
SMART Rate (\$/kWh)	14,317	0.07	0.04	0.00	0.20
Tilt	14,317	29.40	9.48	0	84
Azimuth	14,317	179.86	56.90	0	359
N Inverter	14,317	11.57	13.52	1	78
Annual Generation	14,317	11,289.62	4,428.34	1,547.44	34,383.49
Capacity Factor	14,317	0.14	0.01	0.05	0.17
<i>Panel B: Solar Carve-Out Sample</i>					
Size (kW DC)	25,491	7.96	3.20	1.10	25.00
Total Price (\$)	25,491	33,376.11	13,130.15	5,175.00	138,300.00
Price per kW (\$/kW)	25,491	4,303.56	963.92	2,608.75	7,847.21
Total Subsidy (\$)	25,491	15,169.04	6,435.32	1,471.30	60,419.11
Subsidy per kW (\$/kW)	25,491	1,922.77	336.09	757.12	2,631.60
Net Price (\$)	25,491	18,207.07	8,561.03	796.19	94,096.40
Net Price per kW (\$/kW)	25,491	2,380.79	915.74	183.45	6,300.17
SREC Factor	25,491	0.92	0.10	0.80	1.00
Tilt	25,491	29.39	9.13	0.00	89.00
Azimuth	25,491	181.01	41.31	5.00	356.00
N Inverter	25,491	11.86	14.37	1	280
Capacity Factor	25,491	0.14	0.01	0.05	0.17

Subsidies are present values at an 8% discount rate with 0.5% annual panel degradation. Solar Carve-Out subsidies use the ACP-auction midpoint price path.

Despite the substantial difference in subsidy magnitude, both programs supported installations with similar technical characteristics. Average system sizes were comparable—9.22 kW DC for SMART and 7.96 kW DC for Solar Carve-Out. Capacity factors were nearly identical at 0.14 for both samples, indicating similar performance efficiency across the two program cohorts. The Solar Carve-Out has a sample size of 25,491 whereas SMART has a sample size of 14,317 systems.

4 Empirical Strategy

4.1 The Model

I use a standard regression model to quantify the pass-through of solar subsidies to net installation costs:

$$y_{it} = \alpha + \beta \text{Subsidy}_{it} + X_{it}\gamma + \delta_{u(i)} + \delta_t + \varepsilon_{it}$$

where y_{it} is the net installation price per kW after subsidy (gross price minus subsidy, per kW of DC capacity), Subsidy_{it} is the net present value per kW of the subsidy accumulated over 10 years, X_{it} is a set of controls, $\delta_{u(i)}$ is the fixed effect for the utility $u(i)$ that serves project i , and δ_t is the year fixed effect. Because y is net of the subsidy, the coefficient β equals $-\rho$, where ρ is the pass-through rate: a coefficient of -1 means that a dollar of subsidy lowers the net price by a dollar (the gross price does not change), and a coefficient of -0.76 means the gross price rises by 24 cents.

X_{it} includes panel manufacturer, inverter manufacturer, installer, and city, all of which significantly affect the pricing of a system. Panel and inverter manufacturer fixed effects control for equipment quality, installer fixed effects control for labor costs, and city fixed effects control for demographic characteristics and local price levels. Because SMART rates vary across utility territories, utility fixed effects allow me to identify pass-through from changes in the subsidy within a utility territory, where the compensation structure is the

same. Year fixed effects capture the effect of technology and learning-by-doing on installation costs, as I observe that the average gross installation cost per kW is higher during the Solar Carve-Out regime than during SMART. Thus I identify pass-through from the changes in ACP and auction prices that affect the market price of SRECs and from the changes in the SMART compensation rate as capacity blocks fill up, as well as from differences in generation across roofs.

I estimate the model separately for the SMART and Solar Carve-Out samples, using the net present value of the subsidy at a discount rate of 8%. I do not include utility fixed effects for the Solar Carve-Out sample because SREC prices do not depend on the utility. I also estimated models that include utility fixed effects for the Solar Carve-Out sample. The results are very close to those without them, suggesting that utility territory has minimal effects on pass-through (Appendix Table 11).

Nominal dollars. All prices and subsidies in the baseline are in nominal dollars, for four reasons. First, both are nominal contracts: a SMART rate is fixed in dollars per kWh at approval, SRECs are sold at dollar prices, and a buyer compares the dollar price quoted at purchase with the present value of nominal payments discounted at a nominal rate. Pass-through is a within-period object, the effect of a dollar of subsidy on the price paid on the same date, so there is no economic reason to convert to a common base year. Second, year fixed effects already absorb the economy-wide price level across years. Third, dividing the dependent variable and the regressor by the same time-varying index also rescales the fixed effects over time, so installer, city, and equipment effects that are constant in nominal dollars are no longer constant after deflation, and the coefficient then partly reflects movements in the deflator. Fourth, the consumer price index is not a cost index for solar installation, whose costs follow module, labor, and soft-cost trends that differ from consumer prices, so deflating imposes an arbitrary trend. The choice does matter for SMART, because consumer prices rose by about 22% over the SMART sample while rates were stepping down. Table 6 in Section 5.4 reports the estimates under alternative treatments of the price level and discusses

what they imply.

4.2 Identification and Inference

The regression compares projects that face different subsidies per kW but are otherwise similar in installer, town, equipment, and year. Two sources of variation are exploited, and they differ in how plausibly they are independent of installer pricing decisions. Each can be isolated by a different specification.

Policy variation. Under Solar Carve-Out, the subsidy per kW varies across installation years through the SREC price path and, in January 2017, through the change in the SREC factor. The year fixed effects absorb any level shift common to all projects in a year, so identification comes from how the subsidy differs across projects within a year, which reflects generation potential and SREC factor, and from the interaction between the annual price path and the timing of payments (a project installed in 2016 receives a different sequence of vintage prices than one installed in 2018). Under SMART, the base rate falls as capacity blocks fill within a utility territory, and the value of energy differs by utility and over time. Utility fixed effects hold the compensation schedule constant, so the variation comes from movement along the schedule. Year-month fixed effects (utility-by-year-month for SMART) absorb this variation, and the designs in Section 5.3 isolate its discrete, rule-based part.

Generation variation. Subsidy per kW depends on the electricity a system will produce, so two otherwise identical projects in different locations or with different orientations face different subsidies. This variation exists in both programs and does not depend on the policy calendar. It is arguably the cleanest source, because an installer does not price a north-facing roof differently in order to capture a subsidy. But it is also the source most exposed to measurement error, since PVWatts does not observe shading, and classical errors in its inputs pull the estimate toward one (Section 3.3). Section 5.2 isolates this variation by absorbing the timing of the subsidy and comparing projects of the same installer in the same town and year.

The main threats to a causal reading are the following. First, installation costs fall over time, and block changes and price changes coincide with those declines. Year fixed effects absorb a common trend, but not differences in the pace of cost decline across installers or regions. Second, demand shifts can move both prices and the position on the SMART block schedule, because blocks fill faster when demand is strong. This would bias the coefficient toward zero and thus make pass-through appear lower, not higher. Third, the discount rate enters both the dependent variable and the regressor, so the coefficient mechanically depends on it. I therefore report a range of rates, although for SMART the discount rate only rescales the subsidy (Section 5.4). Fourth, because installers know the subsidy schedule, one cannot separate a pass-through effect from installer pricing that anticipates the schedule. Section 5.2 addresses the first two threats by removing the variation that comes from the timing of the subsidy, and Section 5.3 by using two designs that rely on discrete, rule-based changes in the subsidy: the cut in the SREC factor on January 8, 2017, and the step-downs in SMART rates.

Inference. I cluster standard errors at the installer level. Pricing decisions are made by installers, the same installer appears in many projects and years, and an installer’s pricing errors are likely to be correlated across its projects. There are 385 installers in the Solar Carve-Out sample and 267 in the SMART sample, which is enough clusters for standard asymptotics. Clustering at the utility level is not appropriate: with only four utility groups in the SMART sample the cluster count is too small, and in the Solar Carve-Out sample there is no utility-based policy variation. Appendix Table 12 shows how the standard errors change under alternative clustering choices. The installer-clustered errors are larger than those clustered by town in both samples, and the conclusions do not change. When I compare the two programs, the standard error of the difference treats the two samples as independent. Because the same installers can operate in both programs, this ignores the covariance between the two estimates.

5 Results

5.1 Baseline Estimates

Table 4 presents the estimated solar subsidy pass-through rates, that is, the effect of a \$1/kW increase in the solar subsidy on the net price system owners pay, when system owners value the future cash flows from incentive programs at a discount rate of 8%. Columns (1) and (5) control only for year fixed effects in Panel A (Solar Carve-Out) and for utility and year fixed effects in Panel B (SMART). Columns (2) and (6) add controls for system quality, namely inverter and panel manufacturers. Columns (3) and (7) add installer and city fixed effects, which control for labor costs and local price levels. Columns (4) and (8) include all controls and fixed effects. The estimated pass-through rates for Solar Carve-Out are stable across specifications, ranging from 0.709 to 0.756. For SMART they range from 0.888 to 1.017. For each dollar per kW of subsidy in net present value, system owners capture 71 to 76 cents on average under Solar Carve-Out, and about \$1.02 under SMART in the full specification.

Table 4: Solar Subsidy Pass-through Estimation: Solar Carve-Out and SMART

	Panel A: Solar Carve-Out				Panel B: SMART			
	Dependent Variable: Net Price per kW				Dependent Variable: Net Price per kW			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Variables</i>								
Subsidy per kW	-0.709*** (0.109)	-0.729*** (0.073)	-0.751*** (0.042)	-0.756*** (0.044)	-0.908*** (0.138)	-0.888*** (0.089)	-1.005*** (0.061)	-1.017*** (0.065)
<i>Controls</i>								
Inverter		✓		✓		✓		✓
Manufacturer		✓		✓		✓		✓
Installer			✓	✓			✓	✓
City			✓	✓			✓	✓
Utility FE					✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	25,491	25,491	25,491	25,491	14,317	14,317	14,317	14,317

Clustered (Installer) standard errors in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. Subsidies are discounted at 8% with 0.5% annual panel degradation. SREC subsidies use the ACP to auction midpoint price path.

Across specifications in Panel A, the pass-through estimates without controls for factors that affect system prices, and with controls for only some cost factors, are all close to the full model, which suggests that the estimated pass-through rate is not driven by variation in cost factors that the controls capture. Specifications in Panel B are less similar to one another. Adding manufacturer and inverter controls alone lowers the estimate slightly (0.888), while adding installer and city fixed effects raises it to 1.005, very close to the full model (1.017). The standard errors fall as controls are added, from 0.138 to 0.065 in Panel B, which is consistent with the controls removing cost variation that is unrelated to the subsidy. I take columns (4) and (8) as the preferred models.

I can reject complete pass-through under Solar Carve-Out (the estimate of 0.756 has a standard error of 0.044, so a test of equality with one gives $p < 0.001$) but cannot reject it under SMART (1.017, standard error 0.065, $p = 0.80$). The SMART estimate is not evidence of over-shifting: a 95% confidence interval is roughly 0.89 to 1.14. Both estimates fall within the range documented in prior studies of solar rebate programs. The Solar Carve-Out rate is close to the 78% found by [Pless and van Benthem \(2019\)](#) for host-owned systems enrolled in California’s Solar Initiative (CSI) and the 84% estimated by [Gillingham and Tsvetanov \(2019\)](#) for Connecticut’s rooftop solar rebate program. [Dong, Wiser and Rai \(2018\)](#) also document nearly complete pass-through in the solar industry, likewise in the context of CSI. The baseline estimates, however, use all of the variation in the subsidy, including its movement over time. Section 5.2 shows that much of the gap between the programs comes from that movement, Section 6 discusses what high pass-through implies for market power, and Section 7 discusses why the baseline estimates differ.

5.2 Isolating Cross-roof Variation

The data cannot fully separate incidence from cost co-movement. The SMART base rate is not tied to installation costs: it steps down on a fixed schedule as cumulative capacity fills the blocks, and the incentive also falls when the value of energy rises. Because blocks fill over

time, however, the subsidy declines during the same years in which installation costs fall, and the SREC price path co-moves with market conditions. When subsidies and costs are correlated over time, even without a causal link, the coefficient on the subsidy reflects both how the subsidy is shared and how costs and the subsidy move together. Year fixed effects remove common trends but not differences in their timing across installers. The designs in Section 5.3 compare projects on either side of a rule-based change in the subsidy and so are less exposed to this concern. Fabra and Reguant (2014) report 86% pass-through of emission costs in Spanish electricity markets and show that high pass-through can arise from cost shocks common to all firms, and the same argument means a coefficient near one in this paper could reflect common cost movements. The generation variation across roofs, which does not depend on the policy calendar, is the more reliable source of identification.

Table 5 reports pass-through estimates that progressively remove the variation in the subsidy that comes from its timing. Column (1) is the baseline. Column (2) replaces year fixed effects with year-month fixed effects, interacted with utility for SMART, which absorbs the SREC price path and the timing of SMART block changes. Column (3) adds installer-by-year and city-by-year fixed effects. Column (4) adds the log of system size in kW DC and indicators for five bins of the DC-to-AC ratio. Column (5) instruments the subsidy with the project’s estimated capacity factor in the specification of column (4), so that only the variation in the subsidy that comes from the roof’s orientation, tilt, and location identifies pass-through. The first-stage Wald statistics are 3,661 (Solar Carve-Out) and 1,298 (SMART). Columns (4) and (5) drop 1,988 Solar Carve-Out projects with missing AC capacity.

Adding installer-by-year and city-by-year fixed effects compares projects of the same installer in the same town and year, and controlling for system size and the DC-to-AC ratio removes design choices that affect both generation and price. The Solar Carve-Out estimate rises from 0.76 to 0.86 (standard error 0.04–0.05). The SMART estimate falls from 1.02 to 0.84–0.89 (0.07–0.12). Once the timing of the subsidy is absorbed, the two programs are

Table 5: Pass-through from Cross-roof Variation in Generation

	(1)	(2)	(3)	(4)	(5)
Solar Carve-Out	0.756*** (0.044)	0.799*** (0.046)	0.826*** (0.044)	0.863*** (0.044)	0.856*** (0.050)
SMART	1.017*** (0.065)	0.887*** (0.062)	0.839*** (0.070)	0.847*** (0.073)	0.894*** (0.115)
Difference (SMART – SCO)	0.260*** (0.078)	0.087 (0.077)	0.013 (0.083)	−0.016 (0.085)	0.038 (0.126)
Time fixed effects	Year	Year-month	Year-month	Year-month	Year-month
Installer × year, city × year			✓	✓	✓
System size, DC/AC ratio				✓	✓
Subsidy instrumented with capacity factor					✓

Pass-through is minus the coefficient on subsidy per kW. All columns include installer, inverter, manufacturer, and city fixed effects (and utility fixed effects for SMART), or their interactions as listed. Discount rate 8%. Observations: 25,491 Solar Carve-Out (23,503 in columns 4 and 5) and 14,317 SMART. Standard errors clustered by installer in parentheses. The standard error of the difference treats the two samples as independent. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

no longer statistically different: the gap falls from 0.26 (0.08) to between −0.02 and 0.04 (0.08–0.13). Most of the baseline gap therefore reflects how each subsidy moved over time relative to costs, and pass-through from cross-roof variation is high but incomplete in both programs. I take column (4) as the preferred specification: 0.86 for Solar Carve-Out and 0.85 for SMART.

Three other results favor the cross-roof estimates over the baseline for SMART. First, several SMART estimates that use the timing of the subsidy are above one: 1.55 (0.23) and 1.32 (0.13) in the two lowest quartiles of the subsidy (Table 10) and 1.27 (0.15) at the step-downs in SMART rates (Section 5.3). Demand under SMART is log-concave (Section 6.2), and with log-concave demand, pass-through above one requires marginal cost that falls with volume, whereas pass-through falls, not rises, when statewide volume is high (Section 6.4). Estimates above one are therefore more plausibly a sign that the timing variation is confounded with cost trends than evidence of over-shifting. Second, the baseline SMART estimate is sensitive to the treatment of the price level, and every alternative in Section 5.4 moves it toward the cross-roof estimate. Third, most of the gap between the programs disappears as soon as year-month fixed effects are added (column 2: 0.09, standard error 0.08), before any of the

other controls.

Two caveats apply. First, the cross-roof estimates rely on the variation most exposed to measurement error in generation, and classical error in the PVWatts inputs pulls the estimate toward one (Section 3.3). If anything, then, true pass-through is below the cross-roof estimates rather than above them, which strengthens the conclusion that it is incomplete. Second, the standard error of the difference between the programs treats the two samples as independent (Section 4.2).

5.3 Rule-based Designs

Two further checks use discrete, rule-based changes in the subsidy and are reported in detail in Appendix A.4. A regression discontinuity around the January 2017 cut in the SREC factor (from 1.0 to 0.8) gives Solar Carve-Out pass-through of 0.54 to 0.89 when the pre-deadline rush of completions in December 2016 is excluded (0.71 in the preferred specification) and 0.71 to 1.11 when it is included: similar in magnitude to the baseline, but sensitive to specification. A stacked event study around nine step-downs in SMART rates finds no break in the gross price and pooled pass-through of about 1.27 (standard error 0.15), consistent with near-complete pass-through but imprecise, because the rate is locked at approval while the data record the date the system began operating. A point estimate above one is, as discussed in Section 5.2, more plausibly a sign that the step-downs coincide with falling costs than evidence of over-shifting. Both designs still compare projects a few months apart, so they complement rather than replace the cross-roof estimates.

5.4 Robustness

Price level. The baseline is in nominal dollars, for the reasons given in Section 4. The SMART estimate is, however, sensitive to how price-level changes are treated: it is 1.02 in nominal dollars, 0.91 when the CPI is added as a control, 0.93 with year-by-month fixed effects, and 0.85 in CPI-deflated dollars (Table 6), while the Solar Carve-Out estimate stays

between 0.73 and 0.80. Consumer prices rose by about 22% over the SMART sample while rates were stepping down, so each of these alternatives also removes part of the co-movement between the subsidy and costs discussed in Section 5.2. I therefore read the baseline SMART result as pass-through that is high, between roughly 85 and 100 percent, and not as exactly complete but near. The cross-roof estimate of 0.85 lies at the lower end of this range.

Table 6: Pass-through Under Alternative Treatments of the Price Level

	Solar Carve-Out	SMART
Nominal dollars, year FE (baseline)	0.756*** (0.044)	1.017*** (0.065)
Nominal dollars, year FE, CPI control	0.760*** (0.046)	0.906*** (0.063)
Nominal dollars, year-by-month FE	0.799*** (0.046)	0.929*** (0.062)
CPI-deflated dollars, year FE	0.729*** (0.042)	0.846*** (0.062)
CPI-deflated dollars, year-by-month FE	0.797*** (0.045)	0.883*** (0.062)
Range of the CPI over the sample	234–253	248–302

Full specifications with installer, inverter, manufacturer, and city fixed effects (and utility fixed effects for SMART). Pass-through is minus the coefficient on the subsidy per kW. Discount rate 8%. Clustered (Installer) standard errors in parentheses. Deflated values use the monthly CPI-U and are expressed in dollars of the first half of 2019 (index 254.412). Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

SREC prices. I perform a further robustness check specific to Solar Carve-Out using alternative assumptions about system owners’ expectations of SREC value. An owner with rational expectations would perfectly predict future SREC prices and sell SRECs once a year at that year’s mean price. However, it is hard for an individual to predict the actual price of SRECs, especially in the initial years after the launch of Solar Carve-Out II. The clearinghouse auction price is supposed to be the lower bound, but for a period SRECs consistently traded below it because of oversupply (newly installed capacity quickly surpassed the annual target set by the RPS). The common practice of contracting aggregators or brokers to handle SREC trades makes the actual SREC sale price even more difficult to predict. I therefore consider three ways of predicting SREC prices in addition to actual market prices.

Optimistic customers use the ACP—the theoretical upper bound—as the predicted SREC price. Pessimistic customers use the clearinghouse auction price. Neutral customers use the average of the ACP and the clearinghouse auction price, which is the baseline. A standard 10% transaction fee applies to both optimistic and neutral customers. Pessimistic customers are not subject to any other fees, since the transaction fee is already embedded in the clearinghouse auction price.

Table 7 presents pass-through estimates for Solar Carve-Out projects under different SREC price assumptions. Only the Solar Carve-Out sample is shown, and the results are again very close to each other. The highest estimated pass-through rate comes from using the alternative compliance payment as the SREC price, and the lowest from using auction prices. The ordering is as expected, because the ACP is the upper bound of SREC prices and yields the largest subsidy, while the auction price is the lower bound and yields the smallest, with the range of estimates being 0.723 to 0.782. The estimate using actual market prices, 0.747, is close to the baseline of 0.756. Although I cannot determine owners’ true beliefs about the income stream SRECs will actually generate, the results under alternative SREC prices again demonstrate the robustness of my pass-through estimate for the Solar Carve-Out sample.

Discount rates. Since subsidy calculation requires discounting future cash flows, the choice of discount rate directly affects both the dependent variable (y_{it}) and the independent variable (Subsidy_{it}) in the specification. [De Groote and Verboven \(2019\)](#) find that when subsidies are delivered through electricity production, consumers apply substantial discount rates to future benefits, estimating an average rate of 15% in Belgium—notably higher than my baseline assumption and substantially above the contemporaneous risk-free interest rate for the Euro, which remained below 1% during the study period. To assess the sensitivity of my pass-through estimates to discount rate assumptions, I conduct robustness checks using alternative rates of 3%, 5%, and 15%.

Appendix Table 13 presents pass-through estimates under varying discount rate assump-

Table 7: Solar Subsidy Pass-through Sensitivity with Different SREC Prices

Dependent Variable:	Net Price per kW			
Model:	(1)	(2)	(3)	(4)
	ACP	Auction	Actual Price	Mid
<i>Variables</i>				
Subsidy per kW	−0.782*** (0.039)	−0.723*** (0.050)	−0.747*** (0.046)	−0.756*** (0.044)
Installer	✓	✓	✓	✓
Inverter	✓	✓	✓	✓
Manufacturer	✓	✓	✓	✓
City	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Observations	25,491	25,491	25,491	25,491

Clustered (Installer) standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Subsidies are discounted at 8% with 0.5% annual panel degradation.

tions. In Panel A, the estimated pass-through rate declines from 0.805 at a 3% discount rate to 0.682 at 15%. In Panel B, the estimate is essentially constant, between 1.013 and 1.022. The discount rate mechanically changes the size of the regressor and the dependent variable. For SMART the change is purely mechanical. The subsidy is $r_i g_i A(d)$ (Section 3.4), so changing d multiplies every subsidy by the same factor. The coefficient on the subsidy in the gross price, $1 - \rho$, is divided by that factor, and $(\hat{\rho}_d - 1)A(d)$ is the same at every discount rate. With $A(8\%)/A(3\%) \approx 0.79$, the baseline estimate implies $\hat{\rho}_{3\%} - 1 \approx 0.017 \times 0.79 \approx 0.013$, which is the estimate in the table, and the test of $\rho = 1$ gives the same p -value (about 0.80) at every rate by construction. The SMART results therefore carry no information about how buyers discount the subsidy. Only for Solar Carve-Out, where the price path differs across cohorts, does the discount rate change the relative subsidy of different projects, and at every discount rate the Solar Carve-Out estimate is significantly below one (Appendix Table 14).

6 Discussion

Pless and van Benthem (2019) formalize pass-through over-shifting as a test for market

power: under perfect competition with non-decreasing marginal cost, pass-through cannot exceed one, so more-than-complete pass-through, together with sufficiently convex demand, reveals that firms have market power. This section uses the same framework in the opposite direction. When demand is log-concave, market power pushes pass-through below one, so high pass-through limits how much market power there can be. I first set out the framework, then estimate demand, and then derive the bound for SMART and compare it with reduced-form evidence.

6.1 Framework

[Weyl and Fabinger \(2013\)](#) provide a framework for interpreting the size of pass-through. Consider a symmetric market in which installers' conduct is summarized by a parameter θ , which equals 0 under perfect competition and 1 under monopoly and does not change with the subsidy. The pass-through rate of a per-unit subsidy is

$$\rho = \frac{1}{1 + \frac{\varepsilon_D - \theta}{\varepsilon_S} + \frac{\theta}{\varepsilon_{ms}}}, \quad (1)$$

where ε_D is the absolute value of the demand elasticity, ε_S is the elasticity of supply, and ε_{ms} is the elasticity of the marginal consumer surplus, which measures the curvature of demand. The term $1/\varepsilon_{ms}$ is zero when demand is log-linear, positive when demand is less convex than log-linear (for example, linear), and negative when it is more convex (for example, constant elasticity). [Pless and van Benthem \(2019\)](#) use the same expression, with an additional term for conduct that changes with the subsidy.

Pass-through equals one only when the last two terms in the denominator sum to zero, which happens in four situations: (i) perfect competition with constant marginal cost, (ii) perfect competition with perfectly inelastic demand, (iii) market power with constant marginal cost and log-linear demand, and (iv) market power with demand more convex than log-linear together with rising marginal cost, so that the two forces offset. In situation (iii),

the profit-maximizing markup is a fixed dollar amount that does not respond to the subsidy. Situation (ii) is implausible for residential solar, whose demand responds to price, and (iv) requires two forces to cancel by coincidence. Situations (i) and (iii) are the leading candidates. They have different implications for who gains from the subsidy. The ratio of the gain in consumer surplus to the gain in producer surplus is $I = \rho/[1 - (1 - \theta)\rho]$, which equals $1/\theta$ when $\rho = 1$. Under (i) installers gain nothing. Under (iii) they gain because they sell more systems at a positive markup. A pass-through rate near one therefore does not by itself show that installers capture none of the subsidy.

6.2 Demand

Distinguishing (i) from (iii) requires the curvature of demand. I estimate demand following [Pless and van Benthem \(2019\)](#). The unit is a ZIP code and year, and the outcome is installations per 1,000 residents. Demand is quadratic in the mean net price per watt,

$$q_{zt} = \beta_1 p_{zt} + \beta_2 p_{zt}^2 + X_z \gamma + \delta_{c(z)} + \tau_t + e_{zt}, \quad (2)$$

so that β_2 measures convexity. X_z contains ZIP characteristics from the 2017–2021 American Community Survey (log median household income, the share of adults with a bachelor’s degree, median age, log median home value, and the share of owner-occupied homes), $\delta_{c(z)}$ are county fixed effects, and τ_t are year fixed effects. The net price responds to demand shocks, so I instrument p and p^2 with the mean subsidy per watt and its square and with the average weekly wage in the county’s building-equipment contracting sector (NAICS 2382, from the Quarterly Census of Employment and Wages) and its square. The subsidy shifts the price buyers pay, and the wage shifts installation costs. Neither should affect adoption except through price. This is the functional form that [Pless and van Benthem \(2019\)](#) use, with the same types of instruments, so a difference in curvature between California and Massachusetts is not an artifact of the specification. From the estimates I compute the price elasticity and

the curvature $1/\varepsilon_{ms} = 1 - qq''/q'^2$ at the mean price, with standard errors by the delta method. Because the quadratic in levels imposes a particular shape, I also estimate the same equation with $\log q_{zt}$ as the outcome. In that form the coefficient on p^2 is zero exactly when demand is log-linear, so it is a direct test of situation (iii), and $1/\varepsilon_{ms} = -2\beta_2/(\beta_1 + 2\beta_2 p)^2$. The sample includes ZIP-years with at least one installation and at least 500 residents.

Table 8: Demand Estimates

	Solar Carve-Out	SMART
<i>Panel A: Quadratic demand (Equation 2)</i>		
Net price (β_1)	0.777 (4.256)	2.521 (0.605)
Net price squared (β_2)	-0.124 (0.716)	-0.439 (0.082)
Price elasticity at the mean	0.27 (1.21)	-1.57 (0.42)
Curvature $1/\varepsilon_{ms}$ at the mean	14.9 (56.0)	5.38 (2.77)
<i>Panel B: Log-quadratic demand</i>		
Net price squared	-0.362 (0.459)	-0.446 (0.075)
Price elasticity at the mean	0.79 (1.09)	-2.16 (0.42)
Curvature $1/\varepsilon_{ms}$ at the mean	7.05 (11.35)	2.14 (1.02)
First-stage Wald statistic, p / p^2	20.3 / 16.3	61.2 / 56.3
Mean net price (\$/W)	2.47	3.35
Mean installations per 1,000 residents	1.54	0.91
ZIP-years	2,171	1,715

Two-stage least squares estimates. Panel A: Equation (2). Panel B: the same equation with the log of installations per 1,000 residents as the outcome, so that the coefficient on the squared price is zero under log-linear demand. Instruments: mean subsidy per watt and its square, and the county building-equipment contractor wage and its square. Controls: ACS 2017–2021 ZIP characteristics, county and year fixed effects. $1/\varepsilon_{ms} = 0$ for log-linear demand, > 0 for log-concave demand, and < 0 for log-convex demand. Standard errors clustered by ZIP code in parentheses. Elasticity and curvature standard errors by the delta method. Appendix A.6 reports alternative specifications.

Table 8 reports the estimates. Under SMART, demand slopes down with an elasticity of -1.57 (standard error 0.42) at the mean price of \$3.35 per watt. This is nearly twice the -0.85 that Pless and van Benthem (2019) report for host-owned systems in California at a

mean price of \$3.70 per watt. The difference may reflect a later period in which buyers had more alternatives to buying a system outright, such as leases and power purchase agreements, which are outside my sample, as well as differences in samples and instruments. The squared term is negative and precise, so demand is concave, and the curvature measure is 5.38 (2.77). The log-quadratic form gives the same answer more sharply: the coefficient on the squared price is -0.446 (0.075), which rejects log-linear demand, and the curvature is 2.14 (1.02). Demand under SMART is therefore log-concave, not log-linear and not log-convex, whereas [Pless and van Benthem \(2019\)](#) find convex demand in California with the same functional form. Both forms bend back at low prices: the level quadratic slopes upward below \$2.87 per watt (a quarter of ZIP-years) and the log-quadratic below \$2.62 (13%), so the estimates describe demand around the mean rather than over the whole range. The result holds across specifications. The elasticity ranges from -1.14 to -1.92 and the curvature from 3.4 to 9.8 when I add utility fixed effects, replace county and year fixed effects with county-by-year fixed effects, weight by population, drop ZIP-years with fewer than three installations, or cluster by county ([Appendix A.6](#)). Under Solar Carve-Out, demand is not identified. The subsidy is driven by the statewide certificate price and varies little across ZIP codes within a county and year, and the estimates change sign across specifications. I therefore draw structural conclusions only for SMART.

Two further limitations qualify the SMART result. First, the curvature is measured with limited precision: the quadratic curvature of 5.38 has a t -statistic of about 1.9 and the log-quadratic curvature of 2.14 about 2.1, and the bound on market power below depends on its sign. Second, the sample excludes ZIP-years with no installations, which truncates demand at low quantities, and truncation affects the estimated curvature directly.

6.3 Market Power under SMART

Equation (1) can be written as $1/\rho - 1 = r + \theta/\varepsilon_{ms}$, where $r \geq 0$ is the slope of marginal cost relative to the slope of inverse demand. When demand is log-concave, market power

pushes pass-through below one, so a pass-through rate near one leaves little room for it: $\theta \leq (1/\rho - 1)\varepsilon_{ms}$. At the baseline point estimates ($\rho = 1.017$ and a positive $1/\varepsilon_{ms}$) the bound is zero. Taking sampling uncertainty into account by drawing ρ and $1/\varepsilon_{ms}$ from their estimated distributions, the 95th percentile of the bound is 0.025 with the quadratic demand and 0.060 with the log-quadratic demand, and fewer than 3% of draws imply log-convex demand. Since the Lerner index equals θ/ε_D , these bounds imply a markup of at most about 2–3% of price.

The cross-roof estimate of 0.85 is the more conservative input. With $\rho = 0.847$ (Table 5, column 4), the bound is 0.034 with the quadratic demand and 0.084 with the log-quadratic demand at the point estimates, a markup of 2–4% of price. The upper percentiles are considerably wider: drawing ρ and $1/\varepsilon_{ms}$ as before, the 95th percentile of the bound is 0.15 with the quadratic demand and 0.35 with the log-quadratic demand, which correspond to markups of about 10 and 16 percent of price at the estimated elasticities. Because measurement error in generation pulls the cross-roof estimate toward one (Section 3.3), the true bound could be wider still. The same equation also bounds the supply slope: $r \leq 1/\rho - 1$, which is at most about 0.12 at the lower end of the confidence interval for the baseline ρ and about 0.18 at the cross-roof point estimate.

The SMART evidence therefore points to situation (i) or close to it: a nearly competitive market in which marginal cost is close to flat, so that buyers receive most of the subsidy and installers gain little from it. The bound assumes that conduct does not change with the subsidy and that marginal cost does not fall with volume. Economies of scale on the other hand would loosen it.

The SMART estimate of 1.02, with a 95% confidence interval of about 0.89 to 1.14, is not evidence of over-shifting. The interval includes values below one, and over-shifting requires market power together with log-convex demand (Pless and van Benthem, 2019), whereas the estimated demand is log-concave.

The comparison with Pless and van Benthem (2019) is direct, because the demand spec-

ification is the same. In California, pass-through of 153% to lessees together with convex demand revealed market power in the third-party-owned segment, while host-owned buyers received 78% of the subsidy. In Massachusetts, host-owned buyers receive about 85% of the subsidy under both programs, and log-concave demand turns high pass-through into an upper bound on market power rather than evidence of it. Over-shifting can reveal market power. High pass-through with log-concave demand means the opposite. The two results are not in conflict, since they concern different segments, periods, and states, and the bound here applies to host-owned systems under SMART.

6.4 Supporting Evidence

Table 9 reports three reduced-form checks. First, a rising marginal cost implies lower pass-through when installers are busy. I measure an installer’s workload as the log of its installations in the month of operation and the month before, relative to the installer’s own mean, and statewide workload as the log of monthly installations relative to the mean of that year, and I interact each, standardized, with the subsidy. Second, market power implies higher gross prices where installers face fewer competitors. I regress the gross price per kW on the log number of installers active in the town and year and, separately, on the town-year HHI, for towns with at least ten installations. Because city fixed effects are included, the comparison is within a town over time. Third, if conduct differs across towns, pass-through varies with concentration.

These checks agree with the structural bound for SMART. Pass-through does not fall when an installer is busy (0.044 per standard deviation of workload, standard error 0.023). Statewide workload lowers it by 0.052 (0.022) per standard deviation, from 1.03 at mean volume to about 0.98 one standard deviation above it, which is the small, market-wide supply response that the bound on r allows. Gross prices are lower where more installers are active, by \$66 per kW per log point ($p = 0.05$), or 1.7% of the mean gross price of \$3,796 per kW. This is small, and it may also reflect demand, since installers enter towns where demand is growing.

Table 9: Workload, Competition, and Pass-through

	Solar Carve-Out	SMART
<i>Panel A: Installer workload</i>		
Pass-through at mean workload	0.757 (0.044)	1.019 (0.065)
Change in pass-through per SD of workload	0.010 (0.020)	0.044 (0.023)
<i>Panel B: Statewide workload</i>		
Pass-through at mean workload	0.758 (0.044)	1.027 (0.064)
Change in pass-through per SD of workload	-0.003 (0.029)	-0.052 (0.022)
<i>Panel C: Town-level competition</i>		
Gross price, log number of installers (\$/kW)	42.5 (31.3)	-66.1 (34.0)
Gross price, town HHI per SD (\$/kW)	-95.7 (14.4)	-0.8 (16.3)
Change in pass-through per SD of town HHI	0.052 (0.030)	0.072 (0.043)
Observations, Panels A and B	25,491	14,317
Observations, Panel C	21,783	11,346

Installer workload is the log of the installer's installations in the month of operation and the month before, minus the installer's mean. Statewide workload is the log of all installations in the month, minus the mean for that year. Town competition is measured in town-years with at least ten installations. The HHI is on a 0–10,000 scale. Workload and HHI are standardized within each program, and the pass-through changes are minus the coefficient on their interaction with subsidy per kW. Gross-price regressions control for subsidy per kW. All regressions include the full set of fixed effects (including city and year). Discount rate 8%. Clustered (Installer) standard errors in parentheses.

Pass-through rises with concentration by 0.072 (0.043) per standard deviation, which is not significant at the 5% level. These are correlations rather than causal estimates, because workload and the number of competitors respond to demand. The same pattern appears when SMART pass-through is estimated by tercile of ZIP-level installer concentration: 0.96, 1.04, and 1.06 from the least to the most concentrated tercile, and equality is not rejected ($p = 0.108$, Appendix A.5). If conduct differed across towns, pass-through would be lower where competition is weaker.

6.5 Solar Carve-Out

Without a demand estimate, the Solar Carve-Out results rest on the reduced-form checks, and they do not fit a rising supply curve. Under perfect competition, an estimate of 0.76 would imply a supply elasticity about three times the demand elasticity (about six times at the cross-roof estimate of 0.86), but pass-through does not change with installer workload (0.010, standard error 0.020) or statewide workload (-0.003 , 0.029). The number of competitors in a town has no relation to gross prices, and prices are lower, not higher, in concentrated town-years ($-\$96$ per kW per standard deviation). A likely reason is the Solarize Mass campaigns, in which a town selected a single installer and offered tiered group prices that fell as more households signed up, which makes a town concentrated and cheap at the same time (Clean Energy States Alliance, 2015). For example, Lexington and Bedford selected Astrum Solar in a campaign that ran until June 2014 (Town of Lexington, 2014), and Astrum, as Direct Energy Solar, is the second-largest installer in the 2015 data. The Solar Carve-Out estimate is close to the 0.78 that Pless and van Benthem (2019) find for host-owned systems in California. None of the forces in Equation (1) that I can measure accounts for the shortfall below one. The estimate measures pass-through relative to the present value of the subsidy at an 8% discount rate. If households value uncertain certificate revenue at less than that, the measured rate falls below one even when installers pass on the whole of what buyers value. This points to the size and predictability of the subsidy,

discussed in Section 7.1.

7 Why Did Baseline Pass-through Differ?

The baseline estimates differ between the programs: 0.76 for Solar Carve-Out and about one for SMART. Section 5.2 shows that most of this gap disappears once the timing of the subsidy is absorbed, so the main explanation is how each subsidy moved over time relative to installation costs. This section examines three further candidate explanations for the baseline gap. None of the three can be isolated cleanly, because the programs differ in size, certainty, period, and eligible population at the same time, so the evidence here is descriptive.

7.1 Size

The first explanation is that SMART subsidies are small enough to be absorbed without a supply response, whereas Solar Carve-Out subsidies, worth about 45% of the gross price, strain installers' capacity. The mean subsidy is \$1,923 per kW under Solar Carve-Out and \$566 per kW under SMART, and Section 3 shows that the supports of the two distributions barely overlap. A direct comparison at the same subsidy level is therefore not possible. What I can do is ask whether pass-through falls with the size of the subsidy within each program. Table 10 reports pass-through by quartile of subsidy per kW. Under Solar Carve-Out the estimate is flat, between 0.77 and 0.78 in every quartile, so within that program larger subsidies are not passed through less. Under SMART the estimate falls from 1.55 in the lowest quartile to 1.09 in the highest, which is the direction the size explanation predicts. However, the lowest SMART quartile has a median subsidy of \$127 per kW and includes projects with no subsidy at all, so the estimate is imprecise (a standard error of 0.23), and the gradient may also reflect falling costs as the subsidy shrinks across blocks. Estimates above one in the two lowest quartiles are hard to reconcile with log-concave demand (Section 5.2), which favors the second reading. Taken together, the within-program evidence is mixed: it

does not support the size explanation under Solar Carve-Out and is consistent with it, but confounded, under SMART. The workload test in Section 6.4 points the same way for Solar Carve-Out: if large subsidies strained installers’ capacity, pass-through would be lower when installers are busy, and it is not.

Table 10: Pass-through by Quartile of Subsidy per kW

	Q1	Q2	Q3	Q4
<i>Panel A: Solar Carve-Out</i>				
Median subsidy (\$/kW)	1,528	1,698	2,151	2,305
Pass-through	0.779 (0.073)	0.776 (0.068)	0.780 (0.062)	0.770 (0.059)
<i>Panel B: SMART</i>				
Median subsidy (\$/kW)	127	458	669	947
Pass-through	1.552 (0.228)	1.317 (0.131)	1.189 (0.091)	1.091 (0.076)

Subsidy per kW is interacted with quartile indicators of the same variable in each program, with the full set of fixed effects. Discount rate 8%. Clustered (Installer) standard errors in parentheses.

7.2 Predictability

The second explanation is that fixed SMART rates reduce uncertainty. Buyers who cannot forecast SREC revenue may value it less than its expected present value, and installers cannot raise prices as far. Three observations bear on this. First, the ACP and auction schedules bounded the SREC price, so the range of possible prices was narrower than for an unregulated market, but market prices fell below the auction floor for several years (Figure 1), so the bounds were not binding in practice. Second, the estimate is not sensitive to how one forecasts prices: it ranges from 0.723 (auction price) to 0.782 (ACP) across the three assumptions, and 0.747 with actual market prices (Table 7). If uncertainty about the price path were the main driver, the estimate should be more sensitive to the assumed path. Third, pass-through under Solar Carve-Out falls with the discount rate (0.805 at 3% versus 0.682 at 15%), whereas the SMART estimate cannot respond to it by construction (Section 5.4). The Solar Carve-Out pattern is consistent with buyers applying high discount rates to uncertain,

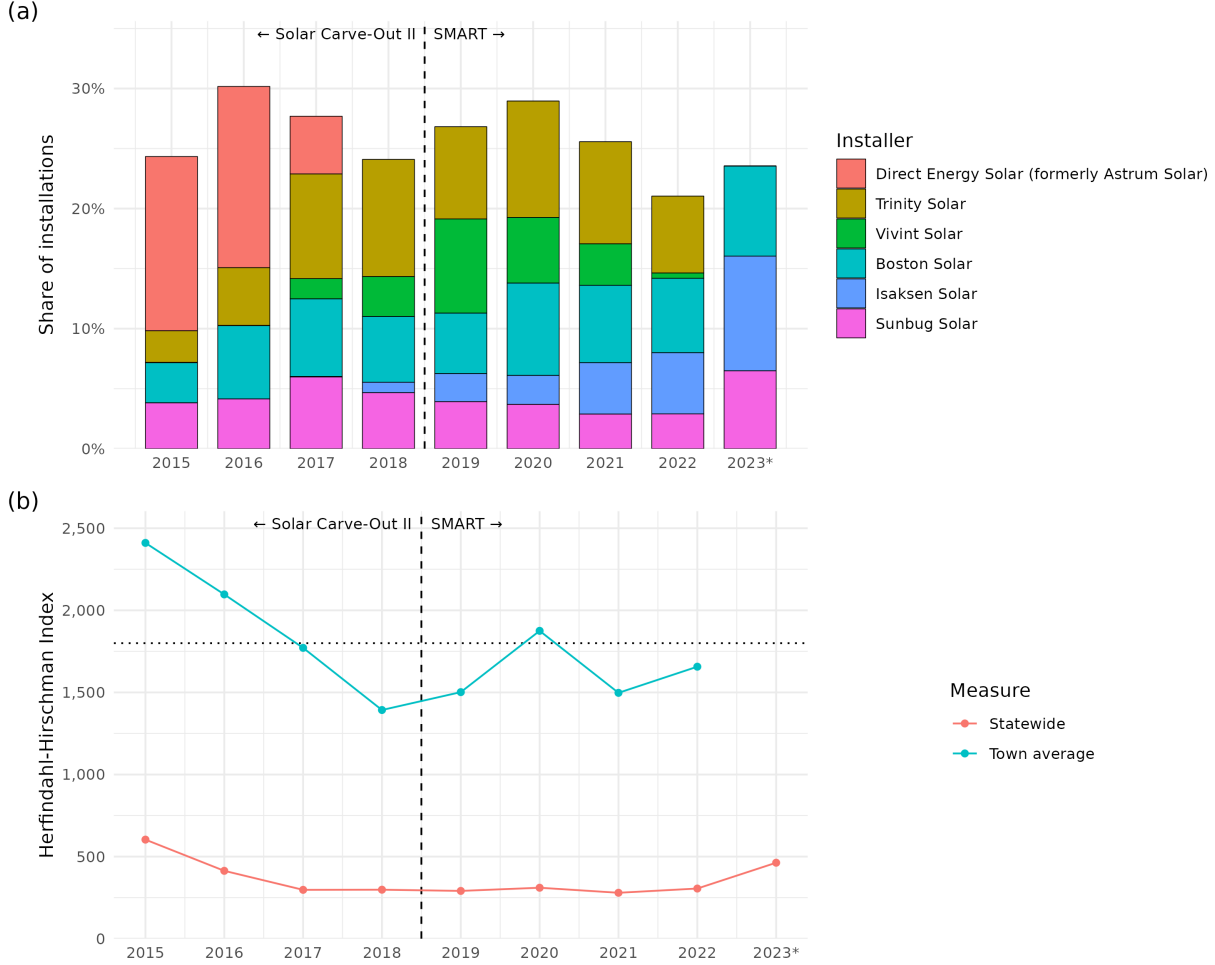


Figure 4: Installer Market Share and Concentration

back-loaded certificate revenue, but it is also partly the mechanical effect of a larger regressor.

7.3 Market Competition

The third explanation concerns competition. Figure 4 shows the concentration of installers over the sample. Panel (a) shows the market share of the four largest installers in each program and Panel (b) shows the Herfindahl–Hirschman index (HHI) at the statewide and the town level.

At the state level, the market is unconcentrated throughout. The statewide HHI is 603 in 2015 and falls to about 300 by 2017, where it stays between 280 and 310 through 2022.

The top four installers' combined share falls from about 40% in 2015 to 23–27% from 2017 on. The number of active installers per year is 184–216 in 2015–2018 and 128–162 in 2019–2022. The figure starts in 2015 because Solar Carve-Out II opened in April 2014 and, in that partial year, only 28 towns have at least 20 installations, so the 2014 town average (above 6,000) is not comparable with later years. Town-level concentration, which is more relevant for a household choosing among installers who quote locally, is higher: the project-weighted town HHI falls from about 2,400 in 2015 to about 1,400 in 2018 and is 1,500–1,900 in the SMART years with no downward trend within the period.

These patterns do not support the idea that the market became less competitive during SMART, nor do they show that it intensified in a way that could explain the jump in pass-through. Most of the fall in concentration happened before SMART started, between 2015 and 2017, when Solar Carve-Out pass-through was already about 0.76. Within each program, pass-through rises only slightly with local concentration, and the change is not significant at the 5% level (Section 6.4 and Appendix A.5). Together with low statewide concentration, this suggests that market power is not the main reason for the gap between the two programs.

8 Conclusion

I estimate the pass-through of two production-based solar subsidies in Massachusetts, using subsidy measures built from each project's expected generation. Baseline pass-through is 76% under the Solar Carve-Out and 102% under SMART, but when only differences in generation across roofs identify pass-through, it is about 85% in both programs. Households capture most of a production-based subsidy, as they do with cash rebates, but not all of it.

Several things matter less than one might expect. The Solar Carve-Out estimates barely move with assumptions about expected SREC prices, and the SMART estimates cannot move with the discount rate by construction. Pass-through does not fall where installers

are more concentrated or when they are busy, and the income gradient within SMART is not robust. The form of the payment, a market-priced certificate or a fixed rate, cannot be separated from the size of the subsidy, and once the timing of the subsidy is absorbed the two forms are passed through at about the same rate, so the data do not show that one design is passed through better than the other.

Three things do matter. The first is competition among installers. Using the test of [Pless and van Benthem \(2019\)](#) in reverse: because demand under SMART is log-concave, high pass-through bounds installer markups at a few percent of price at the point estimates. Unlike California, where convex demand and over-shifting to lessees indicated market power, the market for host-owned systems under SMART appears close to competitive. The second is the timing of the subsidy relative to costs: when the subsidy changes over time together with falling installation costs, part of what looks like pass-through reflects those costs. Using only differences in generation across roofs gives pass-through of about 85% in both programs, with no significant gap between them, and the estimates above one that rely on timing are inconsistent with log-concave demand. The third is measurement: most of the variation in subsidy value is across roofs within the same town, so assigning town or statewide averages would discard it, and errors in measured generation pull pass-through estimates toward one. For policy, the results suggest that in a competitive installer market a production-based subsidy reaches buyers largely, though not fully, and that comparisons of subsidy designs should separate changes in the subsidy from the cost trends that accompany them.

A Additional Results

A.1 Solar Carve-Out with Utility Fixed Effects

Table 11 adds utility fixed effects to the full Solar Carve-Out model. SREC prices do not depend on the utility, so the estimate is essentially unchanged.

Table 11: Solar Carve-Out Pass-through With and Without Utility Fixed Effects

	Baseline	With Utility FE
Subsidy per kW	-0.756*** (0.044)	-0.756*** (0.044)
Installer, inverter, manufacturer, city, year FE	✓	✓
Utility FE		✓
Observations	25,491	25,328

Clustered (Installer) standard errors in parentheses. Discount rate 8%. The number of observations falls because singleton groups are dropped.

A.2 Alternative Clustering

Table 12: Standard Errors Under Alternative Clustering

Clustered by	Installer	City	Utility
Solar Carve-Out	0.756*** (0.044)	0.756*** (0.029)	0.756*** (0.060)
SMART	1.017*** (0.065)	1.017*** (0.046)	1.017*** (0.035)

Pass-through from the full specifications, discount rate 8%. Standard errors in parentheses under each clustering. Utility clustering has only four clusters in the SMART sample and is shown for comparison only. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.3 Discount Rates and Tests of Complete Pass-through

Table 13 reports pass-through at alternative discount rates, and Table 14 the corresponding tests of complete pass-through. For SMART, the estimates differ across discount rates only through the common annuity factor $A(d)$ (Section 5.4): $(\hat{\rho}_d - 1)A(d)$ and the standard error times $A(d)$ are the same at every rate, and so is the p -value.

Table 13: Solar Subsidy Pass-through Sensitivity with Different Discount Rates

	Panel A: Solar Carve-Out				Panel B: SMART			
	Dependent Variable: Net Price per kW				Dependent Variable: Net Price per kW			
	(1) d=3%	(2) d=5%	(3) d=8%	(4) d=15%	(5) d=3%	(6) d=5%	(7) d=8%	(8) d=15%
<i>Variables</i>								
Subsidy per kW	-0.805*** (0.035)	-0.786*** (0.039)	-0.756*** (0.044)	-0.682*** (0.058)	-1.013*** (0.051)	-1.015*** (0.056)	-1.017*** (0.065)	-1.022*** (0.086)
Installer	✓	✓	✓	✓	✓	✓	✓	✓
Inverter	✓	✓	✓	✓	✓	✓	✓	✓
Manufacturer	✓	✓	✓	✓	✓	✓	✓	✓
City	✓	✓	✓	✓	✓	✓	✓	✓
Utility FE					✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	25,491	25,491	25,491	25,491	14,317	14,317	14,317	14,317

Clustered (Installer) standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. 0.5% annual panel degradation. SREC subsidies use the ACP-auction midpoint price path.

Table 14: Test of $\rho = 1$ by Program and Discount Rate

	d=3%	d=5%	d=8%	d=15%
Solar Carve-Out: $\hat{\rho}$	0.805*** (0.035)	0.786*** (0.039)	0.756*** (0.044)	0.682*** (0.058)
p -value	< 0.001	< 0.001	< 0.001	< 0.001
SMART: $\hat{\rho}$	1.013*** (0.051)	1.015*** (0.056)	1.017*** (0.065)	1.022*** (0.086)
p -value	0.795	0.795	0.795	0.796

p -values are for the null $\rho = 1$, using installer-clustered standard errors. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Stars test against zero, and p -values test $\rho = 1$.

A.4 Alternative Identification

The baseline model identifies pass-through from all sources of variation in the subsidy, including trends that could move together with costs. This section uses two designs that rely on discrete, rule-based changes in the subsidy. Both use the commercial operation date as the timing variable and compare projects that differ in the subsidy they face over a short interval.

A.4.1 The January 2017 SREC Factor Cut

On January 8, 2017, the SREC factor for new small systems fell from 1.0 to 0.8 (Section 2). A system that generates the same electricity receives 20% fewer certificates after the cutoff, so the subsidy per kW falls discretely without any change in the installer's costs. Figure 5 plots monthly means by month of operation. The subsidy per kW falls from about \$2,110 in December 2016 to about \$1,610 from February 2017, and every project operating after the cutoff has a factor of 0.8 (the one exception is a single project).

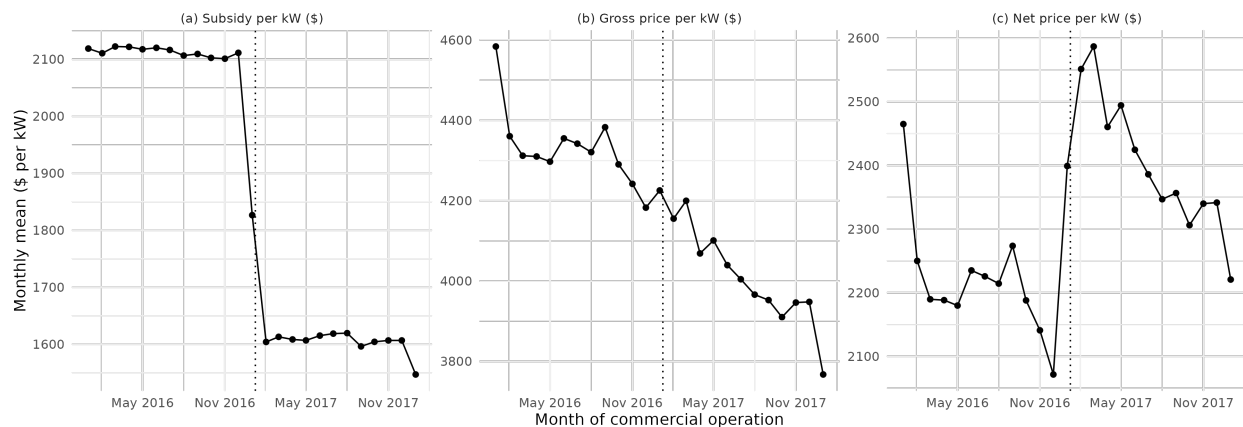


Figure 5: Subsidy, Gross Price, and Net Price by Month of Operation around the SREC Factor Cut

Two features of the raw data complicate the comparison. First, gross prices fall throughout 2016–2017, by roughly \$230 per kW per year, so I control for a linear trend on each side of the cutoff. Second, the number of projects operating in December 2016 (879) is well above

the eleven preceding months (358–647), which suggests a rush to complete before the deadline, and December projects have unusually low gross prices (about \$4,180 per kW, against \$4,240–\$4,380 in the preceding months). I therefore report estimates with and without the two months around the cutoff.

Table 15 reports the discontinuity estimates. The subsidy falls by \$465–\$500 per kW in the baseline measure and \$420–\$435 in the factor-only measure (Panel B), which holds the SREC price path fixed and isolates the change caused by the factor. Excluding December 2016 and January 2017, pass-through estimates lie between 0.54 and 0.89, with a preferred specification (a window of 180 days and the factor-only subsidy) of 0.71 (standard error 0.12). Including those months, the estimates are higher and more variable (0.71 to 1.11), which is consistent with the low December prices widening the apparent jump in net prices. The estimates are noisy at the narrowest bandwidth (a standard error of 0.21–0.25 when the rush is excluded). Overall, this design points to incomplete pass-through under Solar Carve-Out of similar size to the baseline estimate of 0.76, although the range across specifications is wide, and I do not treat any single estimate as more than suggestive.

A.4.2 Step-downs in SMART Rates

The same logic applies to SMART: when a capacity block fills, the base rate for the utility falls by a known amount, and the subsidy per kW falls discretely. I identify step-downs from the monthly modal SMART rate: a step is a month in which the modal rate falls by more than \$0.004 per kWh. Reading the rate this way requires many projects in each month, so I use the two utilities that account for 90% of SMART projects, Eversource East (7,400 projects, 51 months with at least 15 projects) and National Grid (5,747 projects, 46 such months). Unitil has 88 projects and never reaches 15 in a month. Eversource West has 1,082 projects, and its modal rate is noisy in the months with few installations. Adding it gives twelve steps and a pass-through estimate of 1.26 (0.15), almost identical to the estimate below. I keep nine steps that are at least four months from any other step. Each step lowers

Table 15: Discontinuity at the January 2017 SREC Factor Cut

	All projects			Excluding Dec. 2016–Jan. 2017		
	±90 (1)	±180 (2)	±365 (3)	±90 (4)	±180 (5)	±365 (6)
<i>Panel A: Baseline subsidy</i>						
Subsidy per kW (first stage)	−466*** (12)	−478*** (9)	−478*** (8)	−495*** (34)	−497*** (13)	−489*** (10)
Gross price per kW	16 (46)	−65** (33)	−139*** (37)	−117 (104)	−192*** (50)	−227*** (46)
Net price per kW	482*** (48)	413*** (34)	339*** (41)	378*** (114)	305*** (51)	263*** (49)
Pass-through $\hat{\rho}$	1.034*** (0.098)	0.864*** (0.068)	0.708*** (0.079)	0.763*** (0.213)	0.613*** (0.100)	0.537*** (0.096)
Observations	3,206	5,995	11,795	1,722	4,527	10,328
<i>Panel B: Factor-only subsidy</i>						
Subsidy per kW (first stage)	−434*** (11)	−430*** (9)	−424*** (8)	−425*** (34)	−427*** (13)	−422*** (11)
Gross price per kW	16 (46)	−65** (33)	−139*** (37)	−117 (104)	−192*** (50)	−227*** (46)
Net price per kW	482*** (48)	413*** (34)	339*** (41)	378*** (114)	305*** (51)	263*** (49)
Pass-through $\hat{\rho}$	1.110*** (0.105)	0.960*** (0.076)	0.799*** (0.089)	0.888*** (0.246)	0.713*** (0.117)	0.622*** (0.110)
Observations	3,206	5,995	11,795	1,722	4,527	10,328

Each column is a regression on projects within the stated window of January 8, 2017, using commercial operation date as the running variable, with a linear trend on each side of the cutoff and installer, inverter, manufacturer, and city fixed effects. Entries for the first three rows are the estimated jump at the cutoff in dollars per kW. Pass-through is estimated by two-stage least squares with the post-cutoff indicator as instrument for the subsidy. Panel B holds the present-value factor of the SREC price path at its 2016 value, so that the subsidy varies only through the SREC factor and generation. Clustered (Installer) standard errors in parentheses. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

the rate by \$0.011–\$0.020 per kWh. I stack the nine events and estimate the change in subsidy, gross price, and net price in the four months before and three months after each one, relative to the month before the step.

The data record the date a project began operating, while the SMART rate is fixed when the application is approved, which is usually months earlier. The subsidy therefore does not jump at the step but declines gradually across the window (Figure 6, panel a): it is about \$51 per kW above its level in month -1 at four months before, and \$88 below at three months after. If pass-through is complete, the gross price should not respond, and the net price should rise by the amount the subsidy falls. Panel (b) shows that the gross price does not break at the step. Its event-time coefficients lie between $-\$40$ and $+\$26$ per kW, with confidence intervals of roughly $\pm\$40$, and panel (c) shows that the net price rises as the subsidy falls.

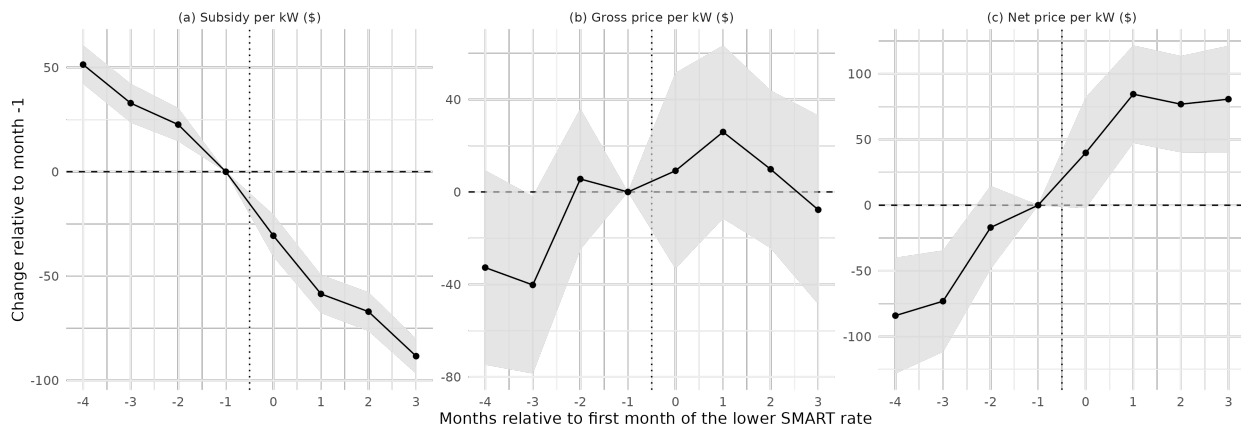


Figure 6: Event Study around SMART Rate Step-downs

Table 16 pools the post-step periods. The subsidy is \$85 per kW lower after a step, the gross price is \$23 higher (a standard error of \$13), and the net price is \$108 higher, which implies pass-through of 1.27 (standard error 0.15) when the post indicator instruments the subsidy. Adding a linear trend in event time leaves the estimate at 1.65 but with a standard error of 1.43, so the design cannot identify pass-through once the gradual decline in the subsidy is absorbed by the trend. Read together, the event study is consistent with pass-

through near or above one at SMART step-downs, in line with the baseline estimate of 1.02, but it is imprecise, it rests on nine events in two utilities, and it cannot rule out that some step-downs coincide with updates to the value of energy.

Table 16: Pooled Event Study around SMART Rate Step-downs

	Estimate
Subsidy per kW, post vs. pre (\$)	-85.2*** (2.4)
Gross price per kW, post vs. pre (\$)	23.2* (12.9)
Net price per kW, post vs. pre (\$)	108.4*** (12.8)
Pass-through, IV (no trend)	1.272*** (0.153)
Pass-through, IV (linear event-time trend)	1.645 (1.435)
Observations	10,793

Stacked sample of nine step-downs in the modal SMART rate (two National Grid, seven Eversource East), with projects from four months before to three months after the first month of the lower rate. Post is the indicator for event time $k \geq 0$. Regressions include event, installer, inverter, manufacturer, and city fixed effects. Clustered (Installer) standard errors in parentheses. Discount rate 8%. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.5 Heterogeneity by Income and Installer Concentration

I examine whether SMART pass-through differs across ZIP codes by median household income and by local installer concentration. Income is the ZIP-level median household income from the American Community Survey (2016–2020 five-year estimates). Concentration is the Herfindahl–Hirschman index (HHI) of installers’ shares of SMART projects in the ZIP code. I divide ZIP codes into terciles on each measure. For each measure I estimate a single regression that interacts the subsidy with tercile indicators and keeps the fixed effects common across groups, and then test whether the three coefficients are equal. Table 17 reports the results.

Table 17: Heterogeneity in SMART Pass-through

	(1) d=3%	(2) d=5%	(3) d=8%	(4) d=15%
<i>Panel A: by ZIP-level median household income</i>				
Low-income tercile	1.074 (0.057)	1.081 (0.062)	1.094 (0.072)	1.125 (0.096)
Middle tercile	1.024 (0.055)	1.026 (0.061)	1.030 (0.070)	1.040 (0.094)
High-income tercile	0.929 (0.054)	0.922 (0.060)	0.910 (0.069)	0.880 (0.092)
<i>p</i> -value, equal pass-through	0.011	0.011	0.011	0.011
<i>Panel B: by local installer concentration (HHI)</i>				
Least concentrated tercile	0.968 (0.062)	0.965 (0.068)	0.959 (0.079)	0.946 (0.105)
Middle tercile	1.032 (0.053)	1.035 (0.059)	1.040 (0.068)	1.054 (0.090)
Most concentrated tercile	1.045 (0.051)	1.050 (0.057)	1.057 (0.065)	1.077 (0.087)
<i>p</i> -value, equal pass-through	0.108	0.108	0.108	0.108
Observations	14,195	14,195	14,195	14,195

Each panel is one regression per discount rate, with the subsidy per kW interacted with tercile indicators. Entries are pass-through rates (minus the coefficient). All regressions include utility, installer, inverter, manufacturer, city, and year fixed effects. Clustered (Installer) standard errors in parentheses. The *p*-value is from a joint Wald test that the three interaction coefficients are equal. It is the same at every discount rate because the discount rate rescales every SMART subsidy by the same factor (Section 5.4). The sample has 14,195 projects, compared with 14,317 in the full SMART sample.

Pass-through falls with income. At the baseline discount rate the estimates are 1.09 in the lowest tercile, 1.03 in the middle tercile, and 0.91 in the highest tercile, and the hypothesis that they are equal is rejected ($p = 0.011$). The pattern is the same at every discount rate, mechanically (Section 5.4). By concentration, the estimates rise slightly from 0.96 in the least concentrated tercile to 1.06 in the most concentrated, the opposite of what market power would predict, and the equality test does not reject ($p = 0.108$). Even if the difference were real, the sign would be the opposite of the market-power story.

Two cautions apply. First, the pattern is not robust to specification. If I estimate the model separately within each tercile, allowing every fixed effect to differ by group, the income gradient disappears: the estimates are 0.97, 1.11, and 0.97 from lowest to highest income tercile, and 1.02, 1.10, and 0.94 across concentration terciles, with no monotone pattern (Figures 7 and 8). The pooled interaction model imposes common installer, city, and equipment effects, which is the appropriate comparison when the question is whether the subsidy coefficient differs given the same cost structure, but the split-sample results show that the gradient depends on this choice, and I do not put weight on it. Second, income and concentration are measured at the ZIP level from SMART records only. Income is an ACS estimate for the ZIP, not the buyer’s own income, and the HHI is computed from the same projects used in the regression, so ZIPs with few projects have mechanically high concentration. Terciles are equal-sized groups of projects and not of ZIP codes.

Taken together, the results are suggestive at most that pass-through is lower for buyers in higher-income areas, and they give no evidence that pass-through is lower where installers are more concentrated. The second finding matters for interpretation: if pass-through under SMART reflected market power, one would expect it to vary with local competition and, since demand is log-concave (Section 6.2), to be lower where competition is weaker. The absence of this gradient is consistent with the view that market power is not the main determinant of pass-through in this market, but the test has limited power, since the HHI measure varies within a range of low concentration statewide (Section 7.3).

Figures 7 and 8 report SMART pass-through when the model is estimated separately within income and concentration terciles, so that every fixed effect can differ by group.

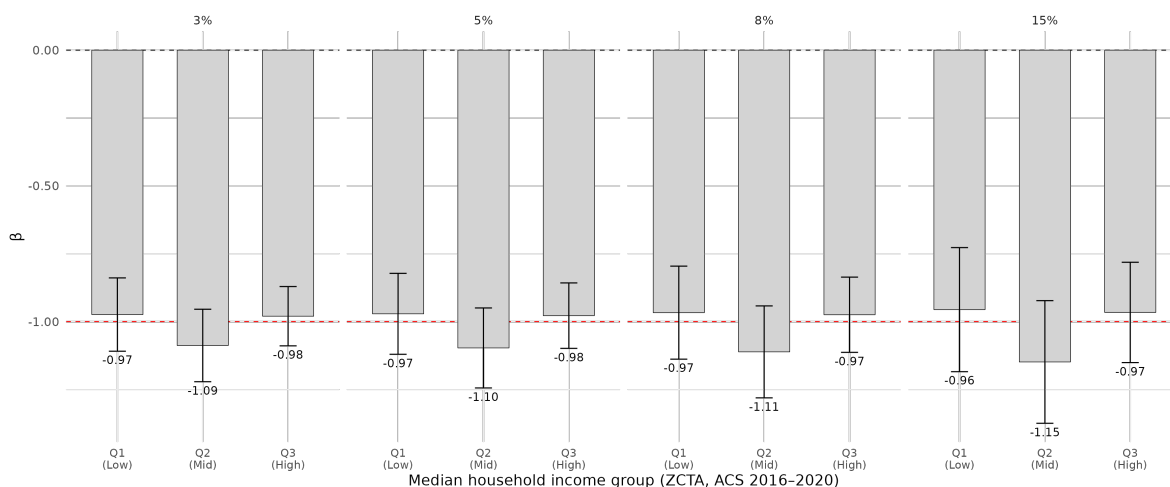


Figure 7: SMART Pass-through by ZIP Income Tercile, Separate Regressions

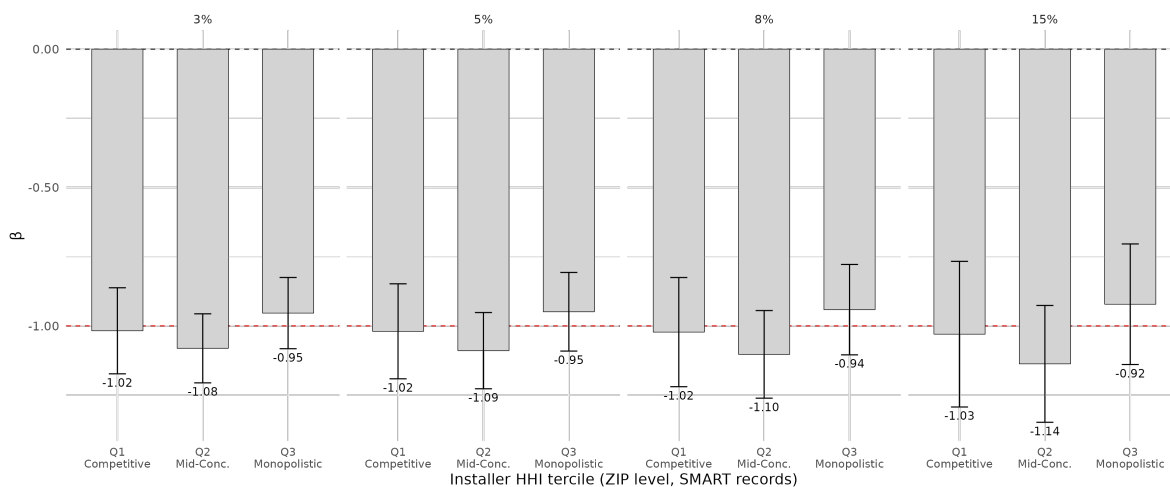


Figure 8: SMART Pass-through by Installer Concentration Tercile, Separate Regressions

A.6 Demand Estimation: Alternative Specifications

Table 18 reports the quadratic demand estimates of Section 6.2 under alternative specifications. With county-by-year fixed effects and only the subsidy instruments, the log-quadratic SMART estimates are almost unchanged: the coefficient on the squared price is -0.425 (0.103) and the curvature is 2.13 (1.12). Under SMART, the price coefficients and the curvature keep their sign in every specification. Under Solar Carve-Out, the estimates are imprecise and change sign, so demand is not identified.

Table 18: Demand Estimates under Alternative Specifications

	Elasticity	β_2	$1/\varepsilon_{ms}$	ZIP-years
<i>Panel A: SMART</i>				
Baseline	-1.57*** (0.42)	-0.439*** (0.082)	5.38* (2.77)	1,715
Utility FE added	-1.92** (0.77)	-0.417*** (0.092)	3.80 (2.65)	1,715
County x year FE, subsidy IV	-1.62*** (0.43)	-0.251*** (0.082)	3.35* (1.73)	1,715
Population-weighted	-1.21*** (0.37)	-0.216*** (0.062)	4.64 (2.97)	1,715
ZIP-years with 3+ installations	-1.14** (0.47)	-0.570*** (0.116)	9.78 (8.39)	1,237
Clustered by county	-1.57*** (0.51)	-0.439*** (0.122)	5.38 (3.55)	1,715
<i>Panel B: Solar Carve-Out</i>				
Baseline	0.27 (1.21)	-0.124 (0.716)	14.86 (56.01)	2,171
Utility FE added	0.30 (1.29)	-0.216 (0.765)	19.58 (97.34)	2,171
County x year FE, subsidy IV	-2.51 (1.61)	-0.202 (0.909)	1.25 (1.43)	2,171
Population-weighted	2.24 (1.40)	-1.001 (0.857)	2.59*** (0.78)	2,171
ZIP-years with 3+ installations	-4.93** (2.46)	5.026** (2.132)	-0.42 (0.91)	1,809
Clustered by county	0.27 (1.12)	-0.124 (0.768)	14.86 (58.00)	2,171

Two-stage least squares estimates of Equation (2). Elasticity and curvature at the mean net price and installation rate of each sample. Baseline: county and year fixed effects, instruments are the subsidy per watt, the county building-equipment contractor wage, and their squares. The county-by-year specification uses only the subsidy instruments, because the wage varies at the county-year level. Standard errors clustered by ZIP code (by county in the last row of each panel) in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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